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Globally Optimal Catalytic Fields – Inverse Design of Abstract Embeddings for Maximum Reaction Rate Acceleration

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Abstract

The search for, and understanding of, good catalysts for chemical reactions is a central issue for chemists. Here, we present first steps towards developing a general computational framework to better support this task. This framework combines efficient, unbiased global optimization techniques with an abstract representation of the catalytic environment, to shrink the search space. To analyze the resulting catalytic embeddings, we employ dimensionality reduction and clustering techniques. This not only provides an inverse design approach to new catalytic embeddings but also illuminates the actual interactions behind catalytic effects. All this is illustrated here with a strictly electrostatic model for the environment and with two versions of a selected example reaction. We close with detailed discussions of future improvements of our framework.

1 Introduction

1.1 Inverse Molecular Design

Computational molecular design usually implies an *inverse* approach, i.e., starting from the desired properties one wants to reach and inferring possible molecular systems that could realize them. However, in chemistry and elsewhere, inverse problems are hard or impossible to solve in practice. A more feasible strategy is *forward* sampling: Starting from many “sensibly” chosen and chemically meaningful systems, the property in question is calculated *directly* – which now is a task in the usual quantum chemistry domain of solving the Schrödinger equation – and then the system coming closest to the desired property value in question is selected, of course *via* an appropriate algorithm and under sufficient automation (see Refs. 1, 2 and references therein). Naive realizations of this forward strategy, for instance deterministic exhaustive enumeration or randomization without any chemical (meta-)heuristics, may hope for serendipity but will be highly inefficient: Chemical compound space^{3,4} is astronomically huge even for small compounds. Hence, a central task of molecular design is to strongly cut down on this vast search space without sacrificing all chances for finding good solutions. In the literature, many different approaches to this problem have been tried, of which we mention some typical examples: Obviously, expert chemical insight is useful,⁵ possibly in combination with simple molecular-orbital models^{6,7} or targeted quantum-chemical calculations.⁸ Another possibility is iterative build-up of complex target structures, with selection steps at each stage.⁹ Both deterministic¹⁰ and non-deterministic^{1,11,12} search has been used to navigate the chemical space of real molecules. Additionally, virtual connections between real molecules have been exploited as shortcuts, in the form of alchemical derivatives and similar methods.^{13,14} Also, data-driven machine-learning (ML) – reviewed also in Refs. 15, 16 – has been used to discover connections between structures and desired properties.^{17,18} Finally, recent applications of (inverse) optimization or ML-based inference tackling more complex properties of molecules also optimized heterogeneous catalysis models

and subsequently their macroscopic properties,¹⁹ general (experimental) chemical reaction conditions,²⁰ and other properties (e.g., for drug discovery or solar cells).²¹

The above can be used to design molecular systems with essentially any desired property. One particularly interesting property, which we also address in the present work, is being an (optimal) catalyst for a given reaction. This has been characterized as one of the “holy grails of chemistry”.²² The most prominent examples of catalysts are of course enzymes,²³ but also transition metal complexes^{9,24} as well as heterogeneous²⁵ catalysts.

1.2 Related Work

Hence, our design target in the present work is catalysis. Of course, it is impossible to provide a comprehensive literature overview on catalysis here, even if narrowed down on analyzing and understanding catalysis mechanisms. Therefore, we only discuss a few exemplary studies here which focused on an *inverse approach* to actual catalyst design and with sufficient similarity to our own work.

Houk et al.^{26,27} introduced “theozymes”, as idealized enzyme active sites represented by a few concrete amino acid side chains, structurally optimized on the computer to achieve optimal transition-state stabilization for the desired target reaction (in a one-point/frame energy scheme, i.e., ignoring gradient information and all other points on the reaction path). Subsequently, this was extended to the design of complete proteins (enzymes) by Mayo²⁸ and Baker,²⁹ and is also often followed by experimental optimizations afterwards.³⁰

Initially focusing on abstract electrostatic surroundings, Sokalski introduced Optimal Catalytic Fields (OCF)^{31,32} and his general Differential Transition State Stabilization (DTSS) concept,³³ both operating in a “differential” scheme (i.e., using a two-point/frame difference method). If electrostatic dominance is found, which is true in many cases, an optimal electrostatic scalar and vector difference field can be constructed: compare Ref. 34 for a concise description of the concept and references therein for similar electrostatics-based approaches,^{35–37} or for a newer application of the differential energetics idea see Ref. 38. In

later work, Sokalski’s abstract OCF and DTSS procedures were linked with concrete active sites of enzymes.^{34,39–41}

Quite recently, Head-Gordon et al. also stressed the importance of the electrostatics for enzyme catalysis and explicitly developed and used MD-sampled averaged (projected) electrostatic fields for the essential breakage and formation of bonds during the reaction.⁴² Moreover, these electrostatic fields were then also used as guidance for (in silico) mutations of de novo designed enzymes mainly for electrostatic transition state stabilization.⁴³

Another line of related work is “Gradient-driven Molecule Construction”,^{1,9,44} aiming at stabilization of otherwise non-stationary points on the potential energy surface (PES). Starting with a general description of an additional energy and interaction term in the Hamiltonian between a fragment and its embedding (“jacket” potential), this work investigated maximal (gradient-based) stabilizations, using different representations: 1.) one explicit partial point charge with continuously varying position, 2.) multiple point charges on pre-fixed positions (motivated by the transition metal complex they were studying) or 3.) an even more fine-grained representation using the additional potential terms on the exchange-correlation functional integration grid.⁴⁴ This approach was recently extended to a *greedy* shell-wise construction framework.⁹ Note that in the present work we employ similar gradient criteria, as one of several fitness ingredients.

1.3 Catalytic Field Model

In our approach to catalyst design presented here, we re-use some conceptual ideas from the literature (mentioned in the previous subsection), but in re-shaped form, as parts of a novel algorithmic strategy. Briefly described, for a single-step reaction to be catalyzed, we construct an abstract surrounding that maximizes an expected catalytic speed-up. This optimization is done with highly efficient non-deterministic global search tools. The final step of translating this abstract but optimized catalytic surrounding into a concrete molecular realization will be addressed in future work; here we focus exclusively on the initial optimization step.

There are no restrictions on the given reaction to be catalyzed. In the remainder of this Article, we will present results for a particular example reaction, but our overall concept is general and should work similarly for any other reaction.

On a boundary layer around this reaction center, enclosing the whole reaction coordinate (from reactants via the transition state (TS) to products), we then introduce what we dub “Globally Optimal Catalysts” (GOCAT in the following). Similar to Sokalski’s catalytic field, this is an abstract representation of a catalyst, but in contrast to Sokalski’s original idea, we understand this GOCAT as a more general, more complex entity that is to be optimized. In an advanced form, it may include the following ingredients, in varying numbers, spatial positions and strengths:

- partial (e.g., point, or multipolar) charges,
- van der Waals interaction centers,
- H-bonding centers (up to also other centers for, e.g., halogen bonding),
- others...

For the present, first proof-of-principle application, we limit ourselves to partial charges. Note that the GOCAT concept is sufficiently extensible to also allow for a transition from the abstract to the concrete: To investigate specific effects in the context of enzymes, more concrete ingredients, for instance capped amino acids, could also be included. This would move the GOCAT concept more towards Houk’s theozymes.

This GOCAT abstraction layer simplifies and shrinks the search space so much that finding an optimal GOCAT can be addressed by non-deterministic global search. For this purpose, we employ Evolutionary Algorithms (EA),^{45–47} because these can be easily mixed with other search paradigms without breaking the overall EA framework, e.g., with local search,^{48,49} with secondary order parameters (i.e. niching)^{48,50,51} or with locally focused exploration.^{52,53} This allows for high gains in search efficiency, by tuning the EA to the problem type at hand,

as we have demonstrated by applying EAs to problems ranging from abstract benchmarks⁵⁴ via cluster structures^{50,55} and molecular design¹² to parameter optimization in reactive force fields.^{56–58}

Additionally, in the EA search for optimal GOCATs, we employ rather elaborate objective functions: They do not only focus on lowering the TS energy and on ensuring practically sufficient reactant affinity and product release, but also include gradient data (not just energies), discretized (in so-called frames) along the whole reaction path. In part, this is necessary to avoid artifacts (lowering the energy at frame N could induce an energy rise at frame $N + M$ that annihilates any catalytic effect), and in part this allows design towards practically useful catalysts (reactant affinity, product release).

Of course, the GOCAT optimization by EA has to use some concrete energy/gradient calculational backend. At this point, our concept is again very general: On a force-field level, a realization would definitely be possible, e.g. employing partial point charges or multipoles as electrostatic GOCAT centers or tunable X-H dummy molecules as H-bond acceptors/donors. However, also a quantum-chemical realization is viable, on any theoretical level (semi-empirical, DFT, ab-initio): The electrostatic GOCAT centers can be conveniently modeled via the QM/MM approach,^{59,60} van-der-Waals centers by tunable rare-gas atoms, etc.¹ See Section 4 for further discussions about the actual model approximations resulting from that. — For the present proof-of-principle demonstration, we have opted for a semi-empirical level of theory, since this gives us raw speed (which is needed for global search in general, and for exploring various conceptual alternatives) while retaining true quantum-chemistry ingredients, indicating that a transfer to higher-level methods is indeed possible.

As we will show below, GOCATs are complex and can exist in many near-equivalent forms. Therefore, we had to employ “big data” techniques to reduce this complexity towards results that can be interpreted and understood. We will briefly introduce only those techniques we

¹ Of course, everything boils down to pure Coulomb interaction at the end.^{61–63} But in the usual interpretation of chemists, one could argue even about separate vdW interactions, H-/halogen-bond interactions etc., which at least in principle could be represented by (fractional) GOCAT entities.

actually need, skipping over the theoretical background and the related literature.

1.4 Disclaimers

Before going into the technical details, a few caveats are in order:

As indicated, we are limiting ourselves here to abstract GOCAT optimization, leaving the necessary translation of such a GOCAT to a real-world catalytic molecular frame for future work. Nevertheless, after complexity-reduction, already such an abstract GOCAT does provide insights into how and why catalytic effects arise.

We are also limiting ourselves to electrostatics (point-charge GOCAT centers) here. For several years, there has been a fierce debate in the literature if this electrostatic effect indeed is the *only* catalytic effect, as forcefully argued by a recent Nobel laureate, Arieh Warshel, in a series of papers,^{64–66} while several other authors equally forcefully insist on the existence of further catalytic effects, e.g., based on enzyme dynamics or a broad range of not-purely-electrostatic enzyme–substrate interactions.^{67–71} We do *not* take sides in this debate: The possibility to argue for “electrostatics only” supports our choice to initially focus only on this ingredient, but the GOCAT concept is broad enough to include other effects.

Homogeneous catalysis frequently involves transition-metal (TM) atoms as crucial ingredients. In later stages of GOCAT development, TM centers may also enter our catalyst design strategy, but we deemed this too complex as a first step.

Last but not least, many real-world catalytic processes involve many elementary reaction steps, frequently arranged in a cyclic fashion and with protons and/or electrons being shifted from one molecular entity to another. Addressing or setting up such multistep catalytic cycles is *not* yet our aim here; instead we merely try to arrive at significantly reduced energy barriers for one reaction step only (even explicitly disfavoring the appearance of intermediate minima). Obviously, one-step catalyst design can be combined with emerging methods for reaction network exploration^{72–77} to arrive at strategies generating and optimizing multistep catalytic cycles. Some of these points will also be discussed in Section 4 again.

The remainder of this article is organized as follows: The EA details including the actual objective function, niching and post-EA clustering procedure is given in Section 2. In the beginning of Section 3, our example reaction is illustrated. Subsequently, results for GOCATs of increasing complexity are illustrated together with discussions of the observed catalytic effects. We will then continue with a further (meta-)discussion of the used GOCAT in Section 4 and end with a conclusion in Section 5.

2 Implementation And Methods

In this first GOCAT illustration we used a pre-defined reaction path. Its structural coordinates remain the same throughout the GOCAT optimization. This path is discretized into a finite number of “frames”. The central GOCAT optimization aim is to change the energies at all of these frames in order to accelerate the reaction, within further restrictions explained below. The GOCAT consists of partial charges placed on a van der Waals surface. This surfaces encloses all frames along the whole reaction path, and the partial charge values and positions are not allowed to vary from frame to frame.

Thus, we have a mixed optimization problem containing two different groups of parameters: Cartesian coordinates on a curved 2D surface and partial charge values. As anticipated (and shown later on), this optimization problem needs an efficient global optimization algorithm having to treat multiple different local minima in an unbiased way. For this purpose, we employed and extended our global optimization framework OGOLEM.^{49,2} The core of this framework is a generation-free⁷⁹ Genetic Algorithm (GA), including options for problem-specific “phenotype” operators,^{50,52,53} niching^{48,51} and further ingredients that increase optimization efficiency. Additionally, OGOLEM features advanced parallelization techniques that allow for failsafe computing distributed across strongly heterogeneous hardware⁸⁰ and for highly adaptive parallel runs that exploit traditionally inaccessible scheduling gaps.⁸¹ Partially,

² At the time of writing, the current work as well as the extensions that are described in Ref. 57 is an own *local* fork of the OGOLEM project. The latter can be found at Ref. 78, for the former contact the authors.

existing OGOLEM algorithm concepts have been re-used or adapted to the current problem. These newly implemented GA details will be briefly described in the following. For general descriptions of our GA framework and its more standard ingredients, we point to former work, see Refs. 49, 51, 57, 80.

In Fig. 1, the general procedure is sketched. In the following, all the different steps are described, in the order of their importance.

General Optimization Steps and Quantum-chemical Theory Levels: For the present GOCAT optimizations, the following key steps were executed; further explanations are given below (in the following, educt is abbreviated as E, transitions state as TS and product as P):

- Part I), **NEB(·)**: Without any GOCAT, the reaction path itself is optimized, at the level of theory used later on. About 20 frames between E and P are defined and constrained during the GA, as well as a common van-der-Waals (vdW) surface enclosing all these 20 frames. **Initialize(·)**: Usually a population (of about $n = 600$ GOCATs) were generated, where the random distribution of charges were weighted by the exposed vdW surface. Each GOCAT consists of a fixed number N_{Ch} of partial charges constrained to be positioned on the vdW surface and constrained to charge values between pre-set minima/maxima throughout.
- Part II): Global GA optimization of the GOCAT population: The genotype dimension is about $4 \cdot N_{\text{Ch}}$ (one charge value and three Cartesian coordinates for each charge). About 10 separate GA runs each for a separate population were executed. **LocOpt(·)**: The final solutions of all runs were *locally* optimized (i.e., in the last GA-iteration for each GOCAT in part II)) with respect to the same objective function as in **Evaluation(·)** (vide infra), using the gradient-free BOBYQA⁸² optimization algorithm in our framework and then accumulated into a common database.
- Part III): This database (of a given N_{Ch}) of all solutions were analyzed, using mainly

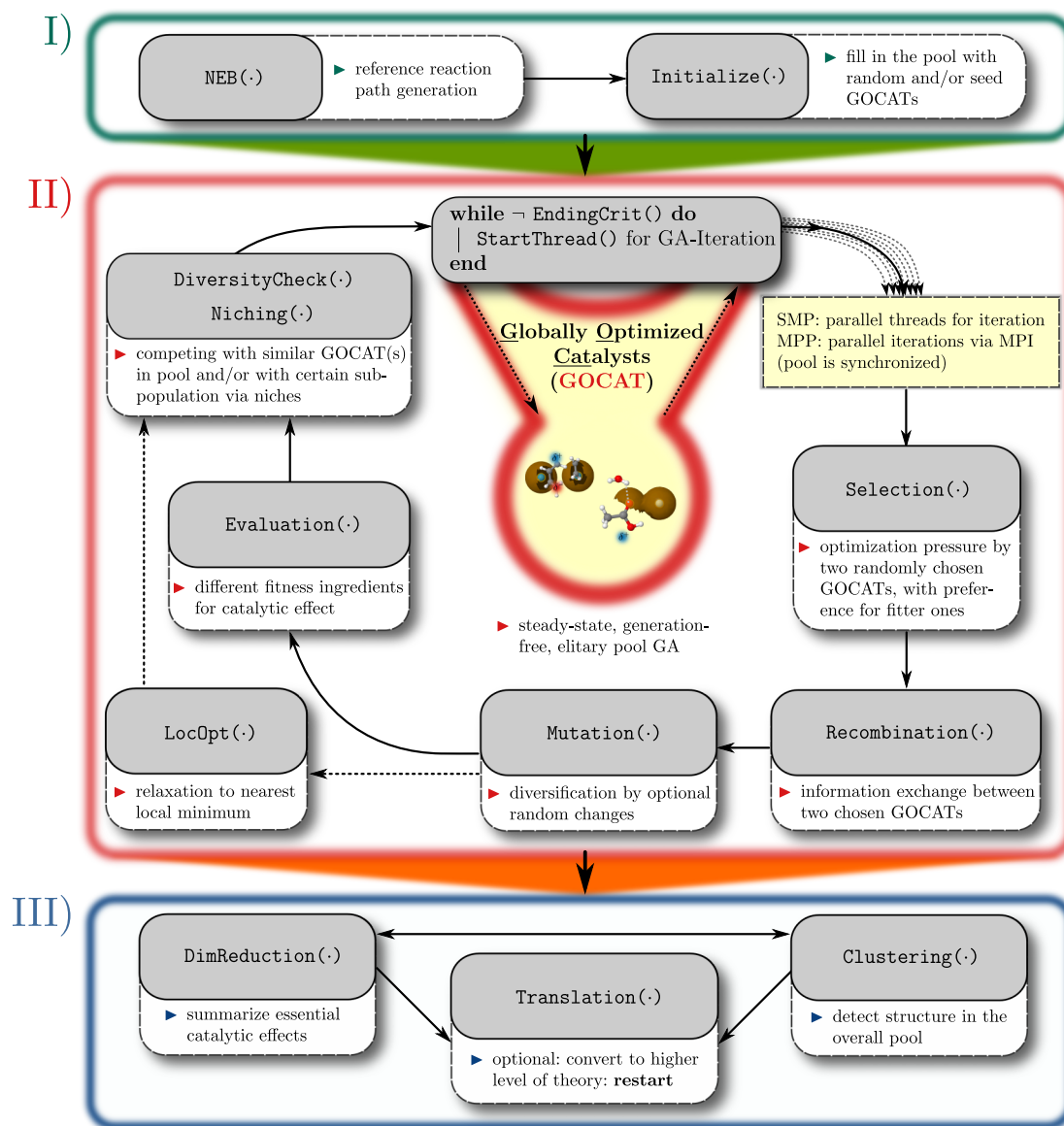


Fig. 1: Overview of the procedure, including: I) reference reaction path creation and initialization, II) GA optimization and III) post-analysis via unsupervised learning/statistics and optional translation to the other level of theory, in which case I) restarts again. All essential algorithm blocks are described in the main text.

cluster analysis methods for identifying meaningful patterns/domains of dominant charge embeddings.

The main observations and discussions presented below will focus on the results *up to and including this last point*.

Our GOCAT framework presented here is intrinsically independent of the level of theory at which the actual calculations are performed. For efficient exploration of the GOCAT search space and to allow a sufficiently wide testing of various changes in algorithm ingredients, we decided to employ a low level of theory in this first GOCAT study. Hence, for all the above steps, we used MOPAC⁸³ with the PM7⁸⁴ Hamiltonian and parameters, and with the (classical) QM/MM coupling scheme available there.⁸⁵ For the exploratory calculations at higher levels of theory mentioned in the next paragraph, we used the ORCA⁸⁶ program suite and mainly DFT calculations with the PBE0⁸⁷ functional and def2-TZVP⁸⁸ basis set.

To explore possible contacts to higher levels of theory, we also added the following steps corresponding to part III) and specifically **Translation**(·), leading to a restart beginning at I) and including the older results as starting GOCATs:

- For several of the most important GOCAT clusters from the cluster analysis, both the best ranks and the nearest neighbors to the (artificial) cluster centers were collected for creating a most diverse input guess for the next steps (seeds).
- To allow for structural changes between the different levels of theory, the vdW surface was re-defined for the higher level of theory and the GOCATs were mapped to this new vdW surface.
- Another GA-based global optimization on this higher level was executed.
- **LocOpt**(·) based on energy, not on the objective function: Additionally, within their given charge embeddings, the reaction frames of the final GOCATs were then locally optimized once more, not with respect to the GOCAT-search fitness function, but

structurally relaxed to locally minimal energies or to reach the next 1st-order saddle point. This caused them to leave the vdW surface and the pre-fixed reaction path slightly, but also to reach converged new stationary points on the new level of theory (with final additional frequency checks to verify the stationary points E, TS and P).

See sections Section 4.1 for further discussions of these higher-level calculations.

Objective Function (Evaluation(.)): All to be optimized properties were accumulated in one aggregate sum as objective function that is to be *minimized* via GA and will also be called fitness function synonymously. After extensive testing, this included finally:

1. $\Delta E^\ddagger = E_{\text{TS}} - E_{\text{E}}$, the (pure electronic) energy barrier between the TS and E should be minimal.
2. The TS should at least be *stabilized* with respect to the reference path without any GOCAT.
3. No new intermediate minima on the reaction paths are allowed, i.e., the reaction profile should remain unimodal.
4. The minima/maximum energy frames (E, TS, P) should not change their positions in coordinate space too much: A fitness penalty was added if the minima/maximum of the GOCAT energy profile moves more than 2 frames off the reference case. However, this penalty does not come into play very often.
5. The gradient norms of the E, TS and P frames should at least be lower than a threshold, i.e., they should approximately retain their character as stationary points.

Note that the gradient restraints item 5. are more important than they may seem at first. Without such gradient restraints, E, TS and P could be arbitrary points, not necessarily stationary ones. Hence, as limiting case, the reaction profile could be transformed into any arbitrary path on the GOCAT-modified PES, even into an equi-potential contour line, with

no barrier at all but also without the defining minimum-energy pathway characteristic (the PES-gradient component perpendicular to the reaction coordinate is not (close to) zero anymore). This could be dubbed over-fitting, since the other requirements item 1. through item 4. are fulfilled perfectly but such a solution would not be a reaction path anymore.

Similarly, item 3. is vital to avoid artifacts. During the GOCAT optimization, each reaction path of a GOCAT solution was analyzed by reading out all maxima and minima and calculating artificial barriers between those, which then enter into the overall fitness. This is necessary as otherwise a new minimum could occur, for instance between E and TS. As a result of that, the effective barrier of a new maximum energy frame minus a minimum energy frame could be higher than the reference barrier. Hence, item 3. effectively enforces a unimodel reaction profile, increasing from E to TS and then decreasing again, such that the final and only barrier included in the globally optimized GOCATs is the one between TS and E.³ Note also, that no stabilization or destabilization (as another possible root of barrier decrease) of the educt frame is enforced in the present objective function definition.

Different weights on the above ingredients were tested, as well as different restraint functions for separate terms. For the results of this paper, the final settings were the following: The barrier, item 1., is highly weighted, since this can be expected to be the main property for creating a catalytic effect. If the TS is not stabilized corresponding to $\Delta E_{\text{TS,stab.}} > 0$, item 2., a penalty of $f(\Delta E_{\text{TS,stab.}}) = p \cdot \Delta E_{\text{TS,stab.}}^2$ was added, with $\Delta E_{\text{TS,stab.}} = E_{\text{TS,GOCAT}} - E_{\text{TS,ref.}}$ and a given weight p . Such a quadratic penalty was also used for all three gradients, item 5., for all $\|\nabla E_{\{\text{E,TS,P}\}}\| > 10 \text{ kcal mol}^{-1} \text{Å}^{-1}$ (explicit weights of the objective function terms are given in the ESI).

The only two non-zero terms within the objective function for the globally and locally optimized solutions shown in this work usually refer to the barrier, item 1., between E

³ Otherwise arbitrary intermediate minima and maxima in the reaction profile could result, turning the single-step reaction into a multistep one, not on purpose but merely as an overfitting artifact. In other systems studied so far, with a non-linear and longer reaction coordinate (hence also containing more frames), this item becomes even more important. So, charting and relating multiple such frames (not just 2 frames) is necessary and is done via this item.

and TS and to the gradient norms, item 5., of the stationary points. All other (penalty) terms, including also the shifted frames, item 4., are already zero after complete optimization. In other words, those latter ingredients help to guide the initial walk through the search space, immediately penalizing non-chemical GOCATs, while a "trade-off" between the former ingredients dominates the endgame and the final fitness ranking.

As in another OGOLEM application,⁵⁷ this insight was exploited to reduce computational expense: In the fitness calculation, the barrier item 1. and gradient item 5. items were calculated before the reaction profile analysis item 3.. If the fitness accumulated from the the barrier item 1. and gradient item 5. items was already worse (higher) than the worst current GOCAT of the GA-population, remaining objective function terms (reaction profile analysis item 3. and the others) were completely neglected and the fitness calculation was stopped, without any influence on our GA progression. Thus, this "Immediate Fallback"⁵⁷ was used as search space reduction technique, saving many QM/MM single-point calculations especially at the end of the GA when the population is already almost converged.

GA Setting: For the GA, we used a mix of different operators: **Recombination(.)**: 1.) A genotype crossover, exchanging both 3D Cartesian information and the charge values between the parents, cutting the genotypes as string of 4D charge entities at several places; 2.) a phenotype operator (similar to the typical phenotype cut-and-splice operator⁸⁹): cutting at 3D planes through the GOCAT and splicing together different parts. Note, that all essential geometric movements by the operators are mapped onto the common vdW surface after each operation again. **Mutation(.)** and random **Initialize(.)**: As mutation, we used also a mix of different operators: 1.) with small probability, a nullary random re-initialization: A packing operator directly on the vdW surface of the reaction, which simply is the random initialization operator for generating the GOCATs in the starting pool. 2.) Usual unary mutation operators for random Cartesian displacements and charge values of a subset of the charges around the current values (within a Gaussian distribution) of the parents. This choice of operators

is the result of an initial benchmarking to increase the general GA efficiency for this kind of optimization, i.e., a greedy brute-force meta-optimization of those GA-settings was done beforehand, but will not be detailed here.

Niching(.): Moreover, we utilized our niching framework⁴⁸ in order to conserve diversity within the population during the GA and to decelerate premature convergence. To this end, a niching version was implemented that is similar to the “dynamic” niching based on the Coulomb matrix which was explained in detail elsewhere.⁵¹ In the present case, the electrostatic potential (ESP), φ_{ESP} , introduced by the charges of each GOCAT serves as molecular descriptor: The net effect (superposition) of any structural composition of partial charges around the reaction path was calculated at the atoms of selected frames of the path (E, TS, P) as:

$$\varphi_{\text{ESP},J} = \sum_i \frac{q_i}{r_{i,J}}, \quad (1)$$

(in atomic units) where i counts the partial charges with values q_i and J counts all separate atoms of the core-frames used with distances $r_{i,J} = |\mathbf{R}_i - \mathbf{R}_J|$ between their position vectors $\mathbf{R}_{\{i,J\}}$. (Incidentally, this is also the combined quantity of the MM atoms (charges) added to the Hamiltonian of the QM part in the QM/MM coupling scheme used in the semi-empirical level of theory.⁸⁵) To define the niches, different procedures like a minimal threshold on a norm of the difference vector between two such ESP-vectors of two individuals were used as well as a vector-element-wise comparison of two individuals with a minimum number of almost identical ESP-values. If such defined similarities are less than a threshold, the individuals were binned to the same niche. As only a maximum number of three individuals per niche were allowed during the GA, new individuals that are supposed to be similar to the ones already in the population just compete with the individuals of that niche as sub-population instead of competing with the complete population. (Thus, at least ≥ 200 niches within the populations of $n = 600$ were always conserved in this concrete case.) Note that this scheme is similar,

but not identical, to a real full-blown cluster analysis based on agglomerative hierarchical clustering described below and used as a post-analysis method.⁴

Selection(·): A rank-based parent selector was applied with a Gaussian distribution centered at the lowest ranks (i.e., the best current individuals), for choosing one parent with a higher probability at the better end of the pool. The other parent was chosen with a uniform distribution. This is standard practice to support both exploration and exploitation.

Reaction Path: **NEB(·):** For the reaction path optimizations we used mainly a nudged elastic band (NEB) optimization (with a climbing image),^{92,93} newly implemented in OGOLEM (and inspired also by the implementation in ASE⁹⁴), together with the FIRE⁹⁵ local optimization algorithm. To our experience, this local optimization algorithm leads to a more fail-proof NEB-optimization than other local optimization algorithms, because of the non-conservative NEB-forces.⁹⁶

Note that of course any scheme for the computation of a minimum energy path (MEP) could have been used as a pre-step to the GOCAT global optimization. But for later (future) flexible reaction paths that are locally relaxed during the GA (cf. the outlook in Section 4.2), NEB will come in handy again (i.e., **NEB(·)** would be part of **Evaluation(·)** itself). Also, we completely aligned each frame of the NEB to each neighbor frame(s) before NEB-distance calculations, essentially removing external degrees of freedom (overall translation and rotation) which was now also published by others elsewhere.⁹⁷

Cluster Analysis: In the domain of Machine Learning (ML) and cheminformatics in general, a lot has already been said about molecular descriptors and similarity metrics between chemical entities (see e.g. Refs. 98–101). In this specific case, the 2-norm of the difference

⁴ One could think of this scheme as greedy, linear-scaling version of single-linkage hierarchical clustering. So, small pre-defined niche-sizes are meaningful because of the *chaining phenomenon* otherwise: Just *two* close individuals would lead to a merge of two niches, without enforcing “compactness” within the whole niche as no other maybe more distant individuals of that niche are considered for the assignment.⁹⁰ We favored this on-the-fly clustering scheme in order not to introduce new serial bottlenecks into our highly efficient generation-free pool-GA.⁷⁹ Yet, in a quite similar context, full-blown cluster analysis at a certain frequency was recently also employed by others.⁹¹

vector of the ESP vectors of Eq. (1), φ , was used as similarity measure, $d(\varphi_n, \varphi_m)$, between two individuals, n , and m

$$||d(\varphi_n, \varphi_m)||_2 = \left(\sum_J (\varphi_n - \varphi_m)^2 \right)^{1/2}. \quad (2)$$

In the case of PM7 level of theory calculations, the reaction path was of C_{3v} symmetry, which was also taken into account during similarity calculations.

Moreover, for cases of almost identical electrostatic embeddings, measured via differences of φ (Eq. (2)), but different structural patterns, the so-called Bag of Bonds (BoB) descriptor was applied (in a slightly adapted version).¹⁰² For short: Based on the Coulomb matrix descriptor,¹⁰³ a vector of concatenated bags, each representing a certain type of “bond”, is formed:

$$\frac{q_i q_j}{|\mathbf{R}_i - \mathbf{R}_j|}, \quad (3)$$

i.e., essentially their attractive or repulsive Coulomb interaction potential energy, but this time including also the partial charge entities (GOCAT). These bags of such terms of Eq. (3) for qualitatively different bonds or interactions types ($\{\text{C-C}, \text{C-N}, \text{Ch-C}, \text{Ch-Ch}, \dots\}$, with Ch as “charge”) are each sorted separately in order to get a permutationally invariant representation (besides the overall translational and rotational invariance as redundant internal coordinate representation) and concatenated together to one (super-)vector for each GOCAT. Note that in the original formulation,¹⁰² the positive nuclear charges are used, whereas in this electrostatic GOCAT embeddings, the core-frames used (E, TS, P) are constrained. So, their atom-pairwise distances as well as their “identities” never change, because in the present case no similarity comparisons of entities throughout the chemical compound space take place like in other ML-approaches for learning properties of *qualitatively different* molecules or molecule classes. Thus, just the partial charges of the charge entities themselves are used in the numerator, while the core atom charges are neglected. Then, in most cases also the Euclidean

norm of the difference of two BoB representations for two individuals was computed.

Both descriptors were used for identity checks (filter procedure): If ESP difference vectors (Eq. (2)) between two individuals in the accumulated database are similar (lower than a threshold of usually $1.0 \text{ kcal mol}^{-1} \text{ e}^{-1}$), the worse individual of such a comparison is erased from the database, if they are also similar with respect to their BoB representation, to test for overall structural (dis-)similarity (compare with Ref. 91, where a similar identity check approach was described).

Clustering(·): For the post-GA analysis part (part III)), we applied the agglomerative hierarchical cluster analysis as unsupervised Machine Learning method implemented in **SCIPY**¹⁰⁴ and **SCIKIT-LEARN**.¹⁰⁵ The general procedure thereby was:

- Each accumulated database for a given GOCAT size (about 6000 individuals) is filtered with respect to fitness and $\Delta E^\ddagger$.
- Filter out further GOCATs by erasing (almost) identical ones (identity check): After local optimization and accumulation, identical solutions might be present as the niching erased identical ones only within the *separate* runs. Thus, a filtered database of about ~ 5000 resulted.
- Full hierarchical clustering using the ESP difference via Eq. (2) as similarity measure was used. Usually, for our purpose, the *average linkage* strategy (unweighted pair group method with averaging, UPGMA) was the best to identify main domains of electrostatic embedding without deforming the clustering space and with the best correlation between composed cluster similarities and the original similarities between the individuals themselves.
- As our main two targets here with this analysis are:¹⁰⁶ 1.) gain some insight into the underlying structure (hypothesis building) and especially 2.) to compress the data for summarization or organization of the results, we used different cutting schemes of the

dendrogram (i.e. final number of clusters), at the end validated mainly through visual inspection of the results.

`DimReduction()`: Additionally, we used standard lower-dimensional projection techniques like Multidimensional Scaling (MDS) and Principal Component Analysis (PCA) to get further insights into the structure of the database.⁹⁰

3 Results and Discussion

As example reaction, we investigate the prototype Menshutkin reaction¹⁰⁷ $\text{H}_3\text{CCl} + \text{H}_3\text{N} \rightarrow \text{Cl}^- + \text{H}_3\text{NCH}_3^+$ as $\text{S}_{\text{N}}2$ reaction with uncharged educt molecules and products with a net charge separation. This reaction usually is endothermic with a high barrier in the gas phase. With increasing polarity of a surrounding environment (solvent), the reaction is more favorable both thermodynamically as well as kinetically due to the stabilization of the charged products and the partial charge separation at the transition state (for some further energetics of this reaction and the impact of solvation and other surroundings, we point to Refs. 108–110 and references therein).

Before we are composing partial charges around the complete reaction coordinate, first of all the (pre-calculated) reference paths are given in Fig. 2. Additional geometric measures are specified in Table 1. For the remainder, the 3 explicitly given frames of the stationary points (given in the insets) are superposed in all of the GOCAT pictures that are following. Here, we will delay the discussion of the details of the reaction path to a later Section, where the geometric differences (and with this the possible Cartesian placements of the embedding and corresponding electrostatic influence) are scrutinized.

3.1 Simplest case: $N_{\text{Ch}} = 1$

In the following, we use the nomenclature: *rn-cm-nx* as full cluster name for the cluster analysis part: *rn* defines the best rank or position of each separate cluster in the complete

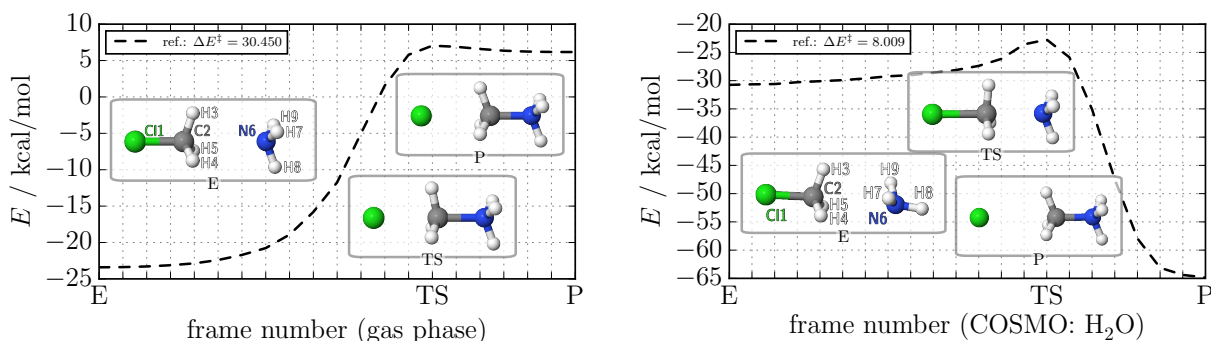


Fig. 2: PM7: Reference gas phase (left) and reference COSMO (H_2O) (right) NEB reaction path, given per frame. Insets show the 3 stationary points. Additional details are compiled in Table 1.

Table 1: PM7: Geometric measures (bond distances and bond angles) of given frames in the insets of Fig. 2. Atom numbering suppressed here (C11, C2, N6), while H3–5 sits at C, H7–9 at N (one H atom used in Table because of symmetry).

Level of Theory	Frame ^a	$r_{\text{CN}} / \text{\AA}$	$r_{\text{CCl}} / \text{\AA}$	$\angle(\text{HCN}) / ^\circ$	$\angle(\text{HCCl}) / ^\circ$
PM7 gas phase ^b	E	3.031	1.775	71.5	108.5
	TS	1.688	2.341	101.5	78.5
	P	1.543	2.589	109.5	70.5
PM7 COSMO: H_2O ^d	E ^c	3.123	1.783	72.0	108.0
	TS	2.146	2.035	84.3	95.7
	P	1.500	3.189	110.6	69.3

^a More precisely: E will be the reaction complex (RC) and P the product complex (PC) below within the surrounding GOCAT, see Section 4.2.

^b Length of total reaction coordinate: 1.963 $\text{\AA}$, TS at 1.737 $\text{\AA}$.

^c NH_3 is rotated off the central attack line.

^d Length of total reaction coordinate: 3.307 $\text{\AA}$, TS at 2.021 $\text{\AA}$.

database, with lower numbers for n (starting at zero) corresponding to lower ($:=$ better) fitness/objective function values. cm is a numbered nickname for the cluster itself and nx denotes the cluster size, i.e., the number of individuals in that cluster.

For getting an impression of how the reaction can already be manipulated by adding just one single charge ($N_{\text{Ch}} = 1$) on the common vdW-surface, Fig. 3 shows a dendrogram for hierarchical clustering of the corresponding database. There in the insets, all electrostatic GOCATs consisting of one partial charge chunked together to clusters are overlain with also overlain selected frames of the reaction for E, TS and P. Each partial charge and also the core

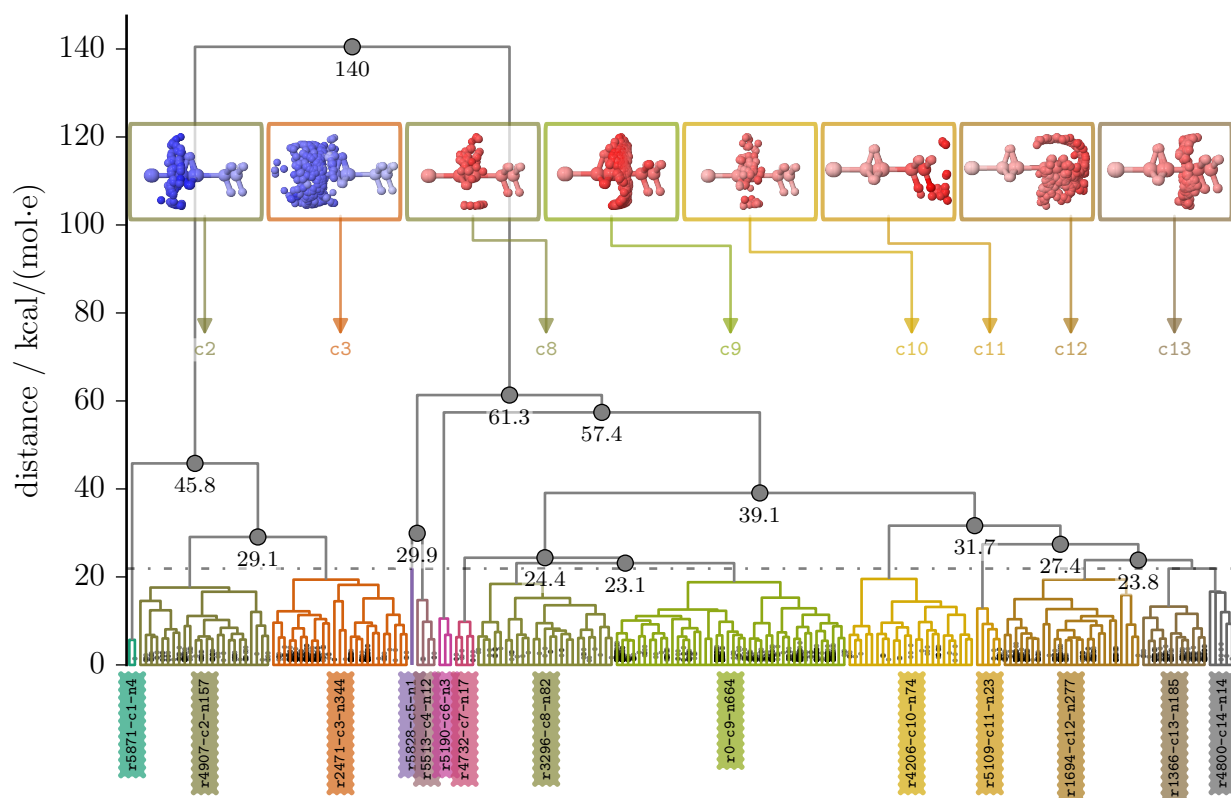


Fig. 3: $N_{\text{Ch}} = 1$ GOCATs (PM7): Dendrogram of the final database with 1857 non-identical individuals using the average linkage strategy. As distance metric, Eq. (2) based on Eq. (1) was used. For illustration purposes the dendrogram was cut to reach 14 different clusters, effectively cutting at a distance of (directly below) $21.89 \text{ kcal mol}^{-1} \text{e}^{-1}$ (dotted line). In the insets, all corresponding overlain GOCATs are shown for some selected bigger clusters, colored between red and blue for negative and positive partial charges and ESP values. This dendrogram is truncated, i.e., the small ellipses at the leaves show additional branching points of the binary tree that are not plotted. The main branching points of the clusters are separately annotated by the numbered dots.

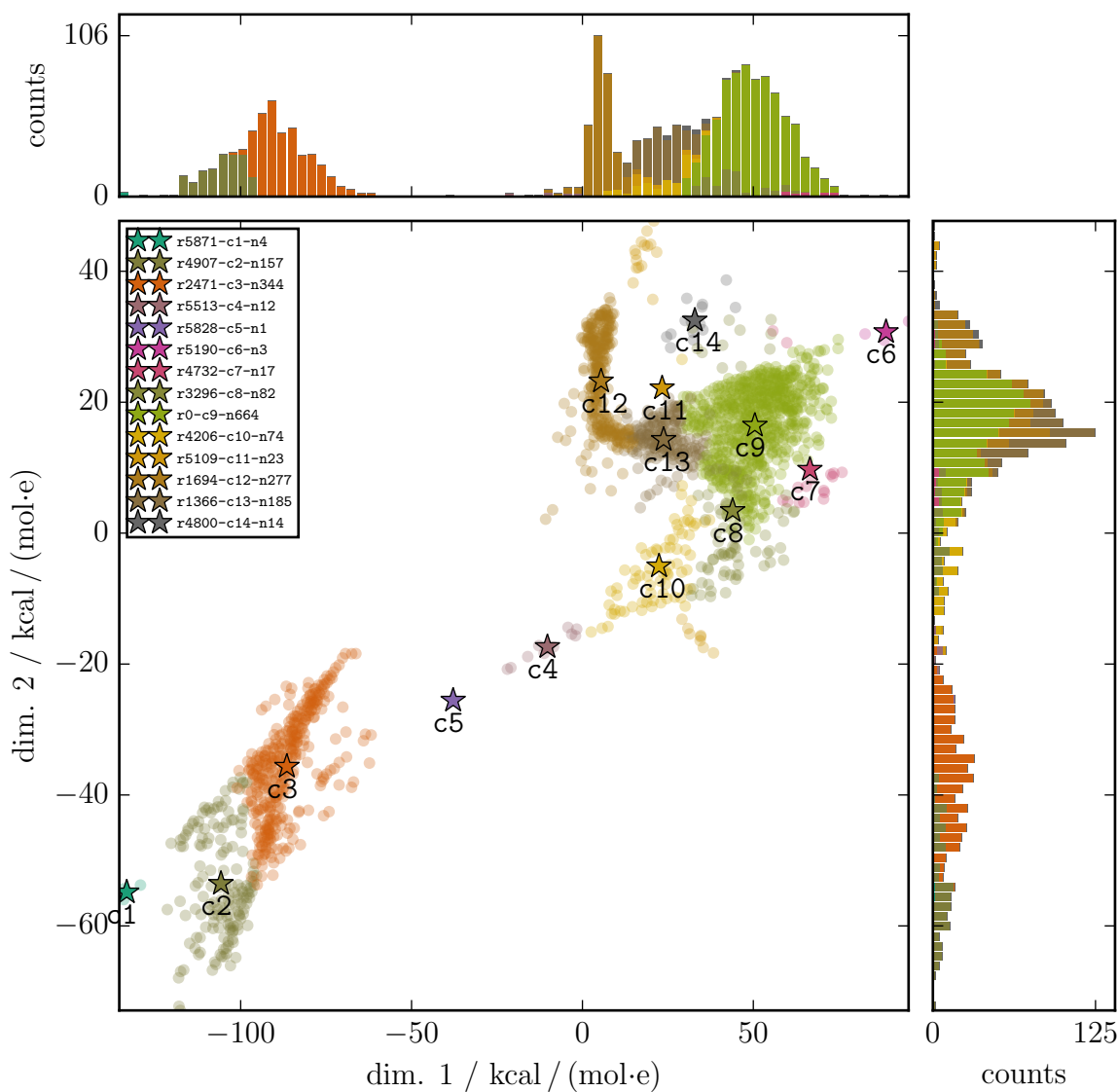


Fig. 4: $N_{Ch} = 1$ GOCATs (PM7): Multidimensional Scaling as 2D projection of the higher dimensional ESP-distance data is shown for illustrating the core cluster regions. Colored stars are the calculated mean 2D-coordinates (centroid) of a cluster. The (unnormalized) stacked histograms illustrate the number of individuals in those clusters.

Menshutkin reaction frames are colored between red and blue for negative and positive partial charges ($[-0.17, +0.17] e$) and ESP-values ($[-36.77, +36.77] \text{ kcal mol}^{-1} e^{-1}$) on those reaction frame atoms, respectively. Because of the sixfold symmetry at PM7 (C_{3v}), each mean ESP vector (for the core atoms) was calculated by starting at the best (lowest) rank of the cluster as a reference and mapping each residual GOCAT of the same cluster to that same symmetry representation before averaging each ESP value at the core atoms. Complementary to this, by showing a 2D MDS, Fig. 4 illustrates the separability of our ESP space in 27 dimensions ($3 \cdot N_{\text{atom}}$ with N_{atom} atoms per frame and 3 frames used: E, TS, P). Individuals with bigger distances (for instance, the biggest distance between the lower left corner vs. the upper right corner) will be mapped to qualitatively different electrostatic embeddings: So, one can divide the database roughly in either positively charged GOCATs or negatively charges ones. The former happen to lie at the lower left corner (big clusters: c2, c3), the latter at the upper right (big clusters: c9, c12, c13) in this 2D projection. This also maps to the highest node (entry-point) of about $140 \text{ kcal mol}^{-1} e^{-1}$ distance in the dendrogram (Fig. 3). In this case, we deliberately stopped at 14 clusters during hierarchical clustering for illustration purposes: Some outliers (small clusters) define their own clusters, while the main regions (funnels) of the final fitness surface are described by a few clusters in this case.

Looking at the overlain GOCAT pictures in the dendrogram, this twofold division of the population leads to the following two funnels: On the one hand, the best catalytic effect (with mean barriers of the cluster of about $\Delta E^\ddagger = 25.4 \text{ kcal mol}^{-1}$) corresponding to the lowest fitness function values, i.e., rank 0 (r0) is found in the main basin of attraction on the fitness surface around cluster c9 (see the GOCAT inset-pictures in Fig. 3 or in Fig. 5 below). As could be expected beforehand, a negative charge (about $-0.11 e$) sitting between the methyl C atom and (attacking) N atom of ammonia is beneficial for the catalytic effect. This stabilizes especially the building of the positive charge (on C and N), resulting finally in the H_3NCH_3^+ product ion. Multiple similar and overlain GOCATs of this kind form a “ring-like structure” in the Cartesian domain. This corresponds to a robust solution with

many qualitatively similar charges in that region.

On the other hand, one positive charge (about $+0.09e$) in cluster **c3** can be placed around the Cl atom, again especially with influence on the product side, when the Cl^- anion forms, resulting in mean barriers about $\Delta E^\ddagger = 26.8 \text{ kcal mol}^{-1}$. A “fitness surface” showing these main basins of attraction (see Figs. **S1** and **S2**), the corresponding statistics (Table **S1**) for this database as well as explicit single GOCATs are given in the ESI.

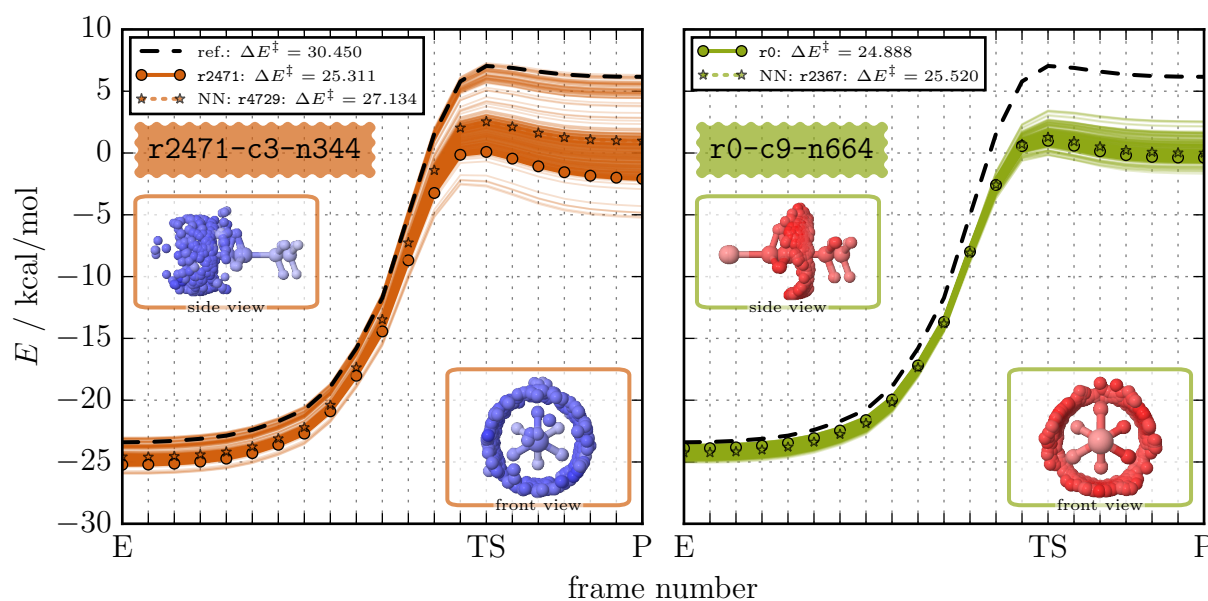


Fig. 5: $N_{\text{Ch}} = 1$ GOCATs (PM7): (Pre-optimized) reaction path for the PM7 reference calculation is shown as well as the path within the electrostatic GOCATs of the 2 main clusters (**c3**, **c9**). The lowest rank of each cluster (circles) and the nearest neighbour (NN) to the mean ESP vector of that cluster (stars) is plotted separately. All other individuals of the cluster are also plotted with thin lines to illustrate the spread. Two images in different perspectives of the overlain individuals are given in the insets.

The resulting reaction barriers within the electrostatics of the GOCAT are shown in Fig. **5** together with the reference barrier in the gas phase. The best rank of the clusters as well as a representative for the cluster center are given explicitly while the rest of the cluster is plotted with thin lines to show the spread and density of the clusters. Note: *Each* GOCAT in those plots leads to a final barrier that is less than the PM7-reference of $\Delta E^\ddagger = 30.45 \text{ kcal mol}^{-1}$ corresponding to a catalytic effect, has gradient norms about $10 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$ and shows a smooth reaction path as defined in the objective function and enforced by the filtering

processes of the database while having qualitatively different influences on the reaction (positively vs. negatively charged at different places). In almost all cases, the stabilization energy (negative energies) relation $\Delta E_{\text{P,stab.}} < \Delta E_{\text{TS,stab.}} \ll \Delta E_{\text{E,stab.}}$ holds: The late TS is stabilized also severely but slightly less than P whereas the neutral E is less influenced leading to an effective (electronic energy) barrier decrease. The remainder of this GOCAT-size reaction profiles of all other 12 clusters are given in the ESI for comparison.

3.2 More charges: $N_{\text{Ch}} = 10$ (non-neutral summed charges)

Going to multiple partial charges as abstract GOCATs increases the complexity of the optimization but also allows to create electrostatic embeddings that address the core-atoms separately, providing some or all of them with individually optimized surroundings, and that may also produce correlated or synergistic effects. We have tried several different N_{Ch} values but present here only the $N_{\text{Ch}} = 10$ size. This is already a converged end-point of a general trend, starting at $N_{\text{Ch}} = 1$, reaching $N_{\text{Ch}} = 3$ (ESI) and not changing qualitatively at $N_{\text{Ch}} = 20$ (not shown). First, without any additional constraints we optimized GOCATs that showed up a very positive or negative overall summed charge. The observations we made can be summed up as:

- More non-redundant GOCATs are possible (bigger search space due to higher complexity/more freedom, and accordingly the already bigger database shown will not be an exhaustive representation of the bigger search space in this case).
- GOCATs happen to lie on a straight line within dimensionality reduction, i.e., the database might even be reduced to *one* (essential) dimension (due to gauge freedom). This provides a hint that one specific qualitative solution (ESP relation on the atoms) is most important for the catalytic effect and that now embeddings with enough freedom or flexibility (of 10 charges) can discover this.
- Very broad ESP range, although the biggest clusters build up at mainly overall neutral

GOCATs and corresponds to broad basins of attraction (funnels) on the fitness surface.

- There is of course no bijectivity of Cartesian placement of partial charges and the resulting ESP at the core atoms anymore (which is also the reason for the additional BOB descriptor mentioned above in the filtering process). This results in different possible Cartesian domains on the vdW surface for similar GOCATs with regard to electrostatics.

A detailed discussion and illustration is given in the ESI. To erase this artificial freedom, in the remainder of this article an additional constraint on overall charge neutrality during GOCAT optimization is added in.

3.3 $N_{\text{Ch}} = 10$ with overall charge neutrality

To eliminate the redundant total GOCAT charge, we repeated the GOCAT optimization for $N_{\text{Ch}} = 10$ of the previous Section 3.2 but now with the additional constraint of overall charge neutrality. This indeed has the desired effect: The dominance of one single dimension in the MDS analysis disappears (ESI: cf. Fig. S22), revealing non-trivial structuring in the other GOCAT characteristics, see Fig. 6, that will be discussed below. Similar plots for hierarchical clustering – from which one cluster is presented in the following – for this case are moved to the ESI.

A stacked histogram of the ESP values of the by far biggest cluster (r0-c11-n3131) of the neutral case, which also includes the best rank found, is illustrated in Fig. 7. There, the biggest cluster with about 3000 GOCATs can be distributed into 3 main domains (Gaussian-like distributions in the histogram). The ESP values at the Cl atom (green ones) almost stay the same upon moving from educt (E-Cl1) to product (P-Cl1) – which then has anionic character – with a positive $\varphi_{\text{ESP,Cl}} = 14 \text{ kcal mol}^{-1} \text{ e}^{-1}$. Also the ESP values at the N-atom (red) can be found at about $\varphi_{\text{ESP,N}} = -4$ to $-7 \text{ kcal mol}^{-1} \text{ e}^{-1}$ from educt to product. The C-atom is even more negatively embedded (blue) and shows a shift from $\varphi_{\text{ESP,C}} = -5$ to about

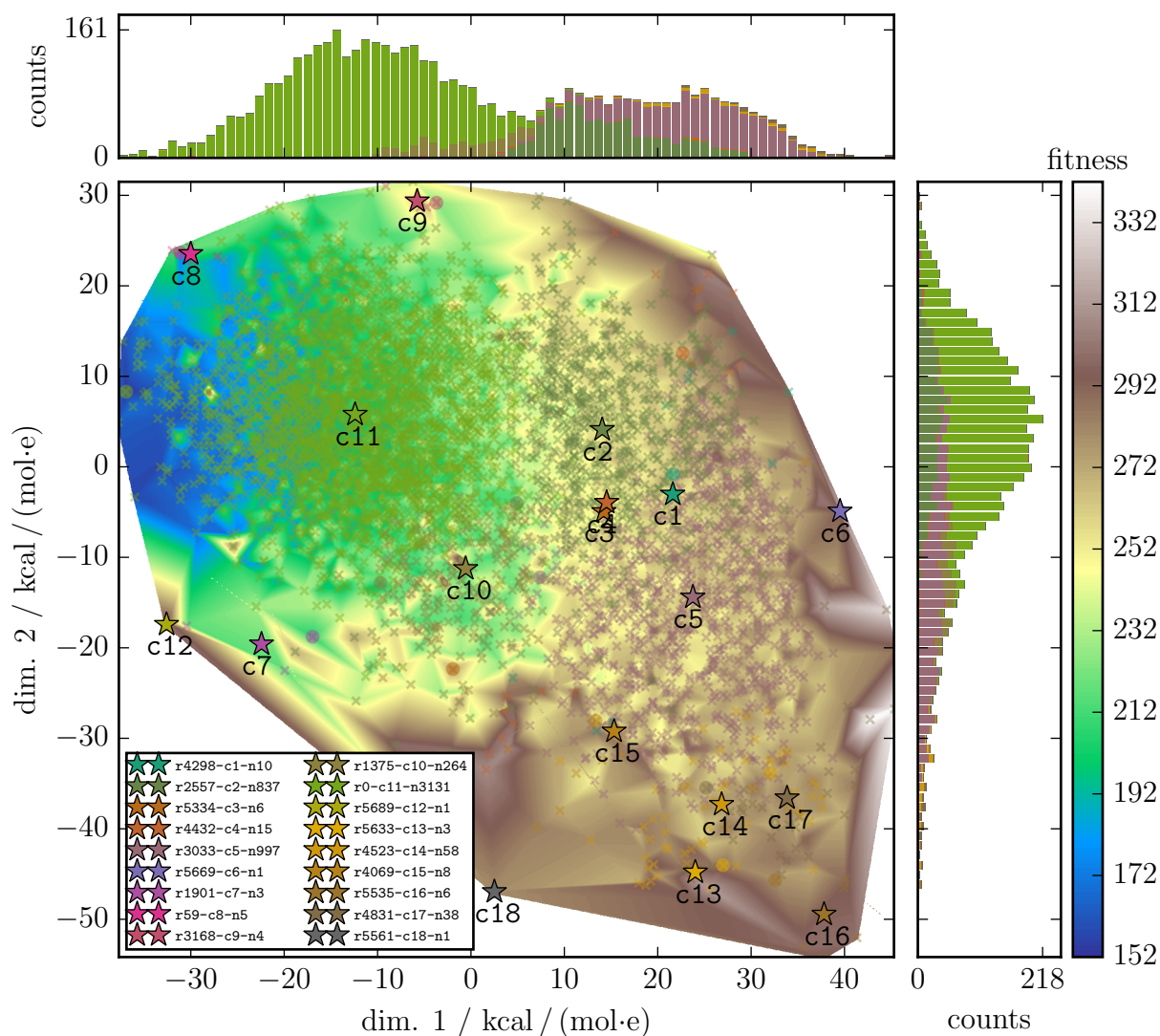


Fig. 6: $N_{\text{Ch}} = 10$ GOCATs (PM7, neutral): MDS 2D projection of the higher dimensional ESP-distance of a database of 5388 non-identical GOCATs. Also, a linearly interpolated fitness surface (color map) is plotted. Besides the cluster means (stars) also the best rank of that cluster (the individual indicated by *rn* in labels of the legend) is plotted as circle. For the other illustration details, compare with Fig. 4.

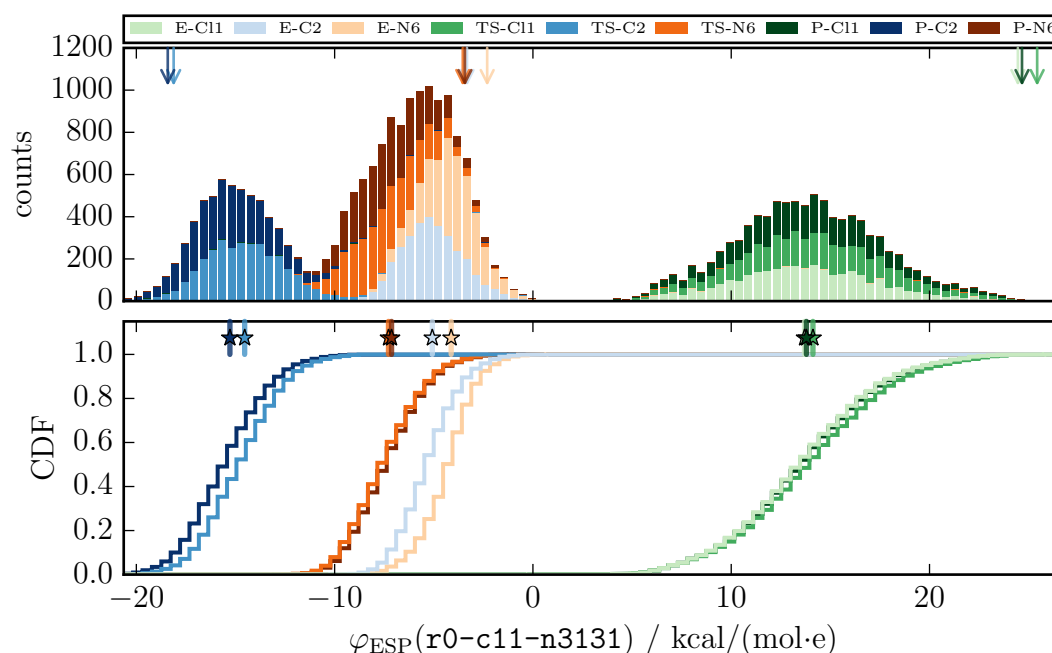


Fig. 7: $N_{\text{Ch}} = 10$ GOCATs (PM7, neutral): Top: Stacked histogram of all the ESP values at the Cl, C and N atoms of the E, TS and P frame of the biggest *neutral* cluster **c11**. Arrows indicate the explicit ESP values of the best rank within their distributions, **r0** (given in Fig. 8), following the coloring of the 9 separate atoms shown in the legend. Below: Cumulative distribution function showing the spread or (if present) skewness; vertical bars with stars are the computed *average* ESP values of that cluster at the corresponding atoms. All values in $\text{kcal mol}^{-1} \text{e}^{-1}$ for E, TS and P frames (standard deviation in parenthesis): Cl: 13.70(3.67), 14.12(3.86), 13.79(3.71); C: $-5.07(1.54)$, $-14.54(2.00)$, $-15.28(2.05)$; N: $-4.12(1.43)$, $-7.28(1.86)$, $-7.13(1.84)$. These are compiled in Table S4 (ESI). Note: This is the only plot type where the color-coding does not follow the hierarchical clustering.

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3 $-15 \text{ kcal mol}^{-1} \text{e}^{-1}$ on average. This comparatively very large shift at the C atom, the small
4 shift at the N atom, the absence of a significant shift at the Cl atom and all average values
5 match nicely with organic chemistry expectations of partial charge changes at these atoms,
6 during this reaction. Hence, these results nicely explain the consistent presence of a catalytic
7 effect here, within the given model approximations. Note that the best rank indicated by the
8 arrows in the upper histogram shows an amplified case with respect to the average values, as
9 its ESP values can be found at the edges of the main distributions of this cluster.
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17 This corresponds to the tendency to find “outlier” solutions (needle-like lowest minima)
18 within the clusters, observed also in the non-neutral case of the last Section (ESI). Thus, Cl
19 is embedded in the best GOCAT more positively up to $\varphi_{\text{ESP,Cl}} = 25 \text{ kcal mol}^{-1} \text{e}^{-1}$, N a little
20 bit less negatively at about $\varphi_{\text{ESP,N}} = -3 \text{ kcal mol}^{-1} \text{e}^{-1}$ and the shift at the C atom from educt
21 to product is even increased from $\varphi_{\text{ESP,C}} = -3$ to $-18 \text{ kcal mol}^{-1} \text{e}^{-1}$. Also all the H-atoms
22 (not shown in this Fig. 7) are embedded highly symmetrically (which is not constrained in
23 general during the GA optimization).
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31 The broad distributions of ESP values in Fig. 7 can be interpreted as a robust solution
32 domain: As long as the ESP is within those ranges, a significant catalytic effect is to be
33 expected, with a final barrier of about $\Delta E^\ddagger = 17.0 \text{ kcal mol}^{-1}$, compare with the corresponding
34 final reaction profiles given in Fig. 8a and others in the ESI in Fig. S31 for all the GOCATs.
35 (Of course, these reaction paths are similar to Fig. S23 for the biggest non-neutral cluster; but
36 because of the charge neutrality constraint here, we obtain a higher resolution of this search
37 space region). But as already mentioned above, with the increased complexity of $N_{\text{Ch}} = 10$
38 GOCATs, several also less symmetric or less extreme solutions are possible that are also (near)
39 local minima within this cluster. A very specific (“outlier”) example is the best solution found,
40 $\mathbf{r}_0$, that is also illustrated in Fig. 8. This is one of many possible Cartesian placements of
41 partial charges that achieve those ESP values at the main reaction atoms (Cl, N, C), but even
42 more amplified as discussed above. Notice the high symmetry of the Cartesian placements
43 themselves (which is not strictly necessary in very many other solutions).
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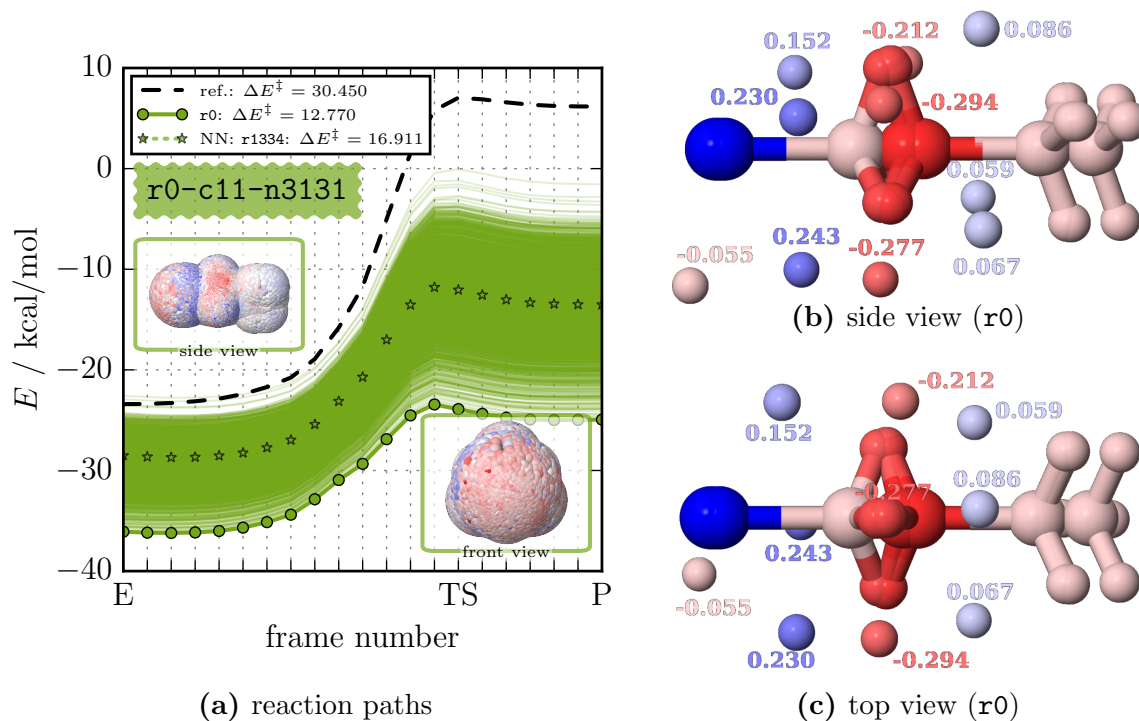


Fig. 8: $N_{\text{Ch}} = 10$ GOCATs (PM7, neutral): In **a**, reaction paths of the biggest cluster (c11) are shown. For illustration details, see Fig. 5. Note, that the overlain GOCATs in the insets of the plots are misleading, as almost a continuous embedding is shown as superposition of, e.g., each $3131 \cdot 10$ charges in the c11 case. In **b**, **c**, two different views of r0 (of c11) with values given for the partial charges. Both partial charges and the atoms of the selected core frames (E, TS, P) are colored red/blue in the ranges $[-0.537, +0.537]$ e for charges and $[-23.578, +23.578]$ kcal mol $^{-1}$ e $^{-1}$ for ESP values. More views are given in the ESI (Fig. S33).

As many other, higher-rank GOCATs, also this lowest-rank one can be interpreted quite easily: The dipole moment (contact ion pair) is nicely reproduced by the GOCAT embedding of the 3 positive (near Cl atom) and the 3 negative partial charges (near C atom in the middle), respectively. Moreover, the positive charges between the attacking NH_3 group and attacked C atom facilitate the nucleophilic attack of the N-atom on the slightly negative C atom, which is then fully stabilized (negatively embedded) at the P-side when the cation has formed (thus the shift of φ_{ESP} of C from E to P in Fig. 7). The remaining single charge of low value sitting at the Cl atom seems to be “mere noise”, i.e., in this $N_{\text{Ch}} = 10$ case it has to be placed somewhere, but in principle it could also vanish, i.e. set to 0.0 e during optimization automatically. Thus, this shown GOCAT is very symmetric, but still not “perfect”, as at least this (tiny) 10th charge and also partially the positions and values of the other charges slightly break the symmetry.

Following this interpretation (and intuition created), one could think of even more symmetric final solutions, maybe the “real” global minimum of this search space, which would render the best rank shown above a prematurely converged one.

The “outlier” quality of the best rank (**r0**) discussed above partially arises because for optimal fitness a non-trivial compromise between the most important fitness ingredients has to be found: between the lowest barriers possible and the smallest gradient norm penalties. There is a clear overall trend on this fitness surface, see Fig. 6 and/or Fig. S30 (ESI), but of the main cluster shown (**c11**) just a tiny fraction is found in the best (“upper left”) lowest fitness region. Looking especially at the best GOCATs there, all these show again a certain symmetry, where the mirror plane is the most important element. With that symmetry, it is also possible to reach absolute higher electrostatic potential values at the core atoms, i.e., the range between negative Cl atom embedding and positive N and especially C atom embeddings increases. With partial asymmetry, gradients (of e.g. differently embedded H atoms) increase and thus penalize most of the asymmetric GOCATs with large charge/ESP values. Since our GA has no built-in preference for symmetry, asymmetric GOCATs are much more likely

than symmetric ones. Therefore, most individuals in the c11 cluster do not make it into the upper left, lowest-fitness corner. However, as discussed above, also with less (or no apparent) symmetries of the GOCAT, the main catalytic effect is already established, though to a smaller extent. One such example – the nearest neighbor to the computed cluster mean (and thus very near the centers of the distributions in Fig. 7) is given in the ESI (see Fig. S34).

3.4 $N_{\text{Ch}} = 10$ stabilizing COSMO path

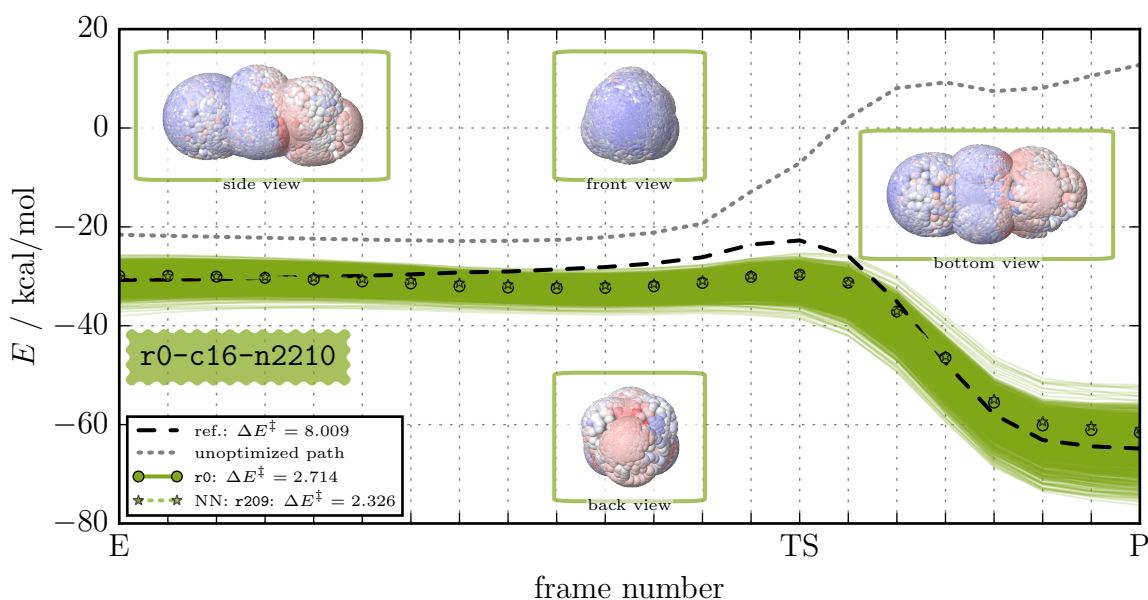


Fig. 9: $N_{\text{Ch}} = 10$ GOCATs (PM7, neutral, COSMO): Reaction paths of the biggest GOCAT cluster (c16) starting with the COSMO structures, but stabilized solely by partial charges (gas phase). For illustration details, see Fig. 5. The “unoptimized path” are the energies of the COSMO NEB structures calculated in the gas phase without any GOCAT.

The strongest restriction of the current electrostatic GOCAT model is the static pre-optimized reaction path. This forces all stationary points to retain their character, despite the changing electrostatic embedding. We further discuss the influence of this assumption and how to go beyond it in the discussion and outlook section below. But even with retaining this restriction here, we still have the freedom to select any other pre-defined reaction path as input. We illustrate the possible benefits of other path choices for the same reaction here, by

selecting a NEB path with H₂O as solvent, treated by the implicit solvent model COSMO.¹¹¹ As it is well known for this reaction in a protic solvent, on such a path the TS is shifted towards the E frame, the barrier is lowered, and the reaction is overall exothermic – in concordance with Hammond’s postulate,¹¹² i.e., a late TS in the gas-phase is shifted to the educt with increased stabilization due to the surrounding. With a slightly changed objective function, the target now is to stabilize this COSMO path – not with the COSMO implicit solvent but by our electrostatic GOCAT. This allows us to compare not only GOCAT-embedded reaction profiles to the gas-phase profile for the same path (as done above), but also to compare the GOCAT-embedding to the COSMO embedding. Moreover, this will also lead to the discussion of several other model assumptions.

So, the objective function now did not include the maximal barrier decrease, item 1., and not the TS stabilization, item 2., of Section 2 but instead tried to reach the COSMO energies themselves for each frame, and the gradient norms on the stationary points (E, TS, P) were minimized as well. (Note, however, that further tests indicated that stepping back from this setting to the original one in Section 2 leads to qualitatively very similar results as reported here.) The other ingredients were not changed (explicit weights of the terms are given in the ESI again). Additionally, as in Section 3.3, the final summed charge of the GOCAT is constrained to zero.

With this COSMO path and these fitness ingredients, yet another series of global and local optimizations was performed, again collecting the results into a common database and analyzing the latter with the same tools already used above. For the biggest cluster in that database, r0-c16-n2210, the final reaction profiles are illustrated in Fig. 9 (see the ESI for the clustering plots). In general, the optimization target to reach a stabilized COSMO path (with respect to the gradients of the stationary points as well as the energies) was successful: The overall energy trend of the COSMO reaction could be reached with just 10 explicit partial charges on a vdW surface. The GOCATs are able to stabilize the differing geometric features of the structures of the COSMO case and are able to reproduce the strong stabilization of

the ionic products to reach $E_P < E_E$. At the same time (and as “side effect”, as this was not an explicit optimization target in this case) the reaction barrier decreases significantly. Still, there also are a few small deviations between the COSMO and the GOCAT energy profiles; they will be discussed and explained below.

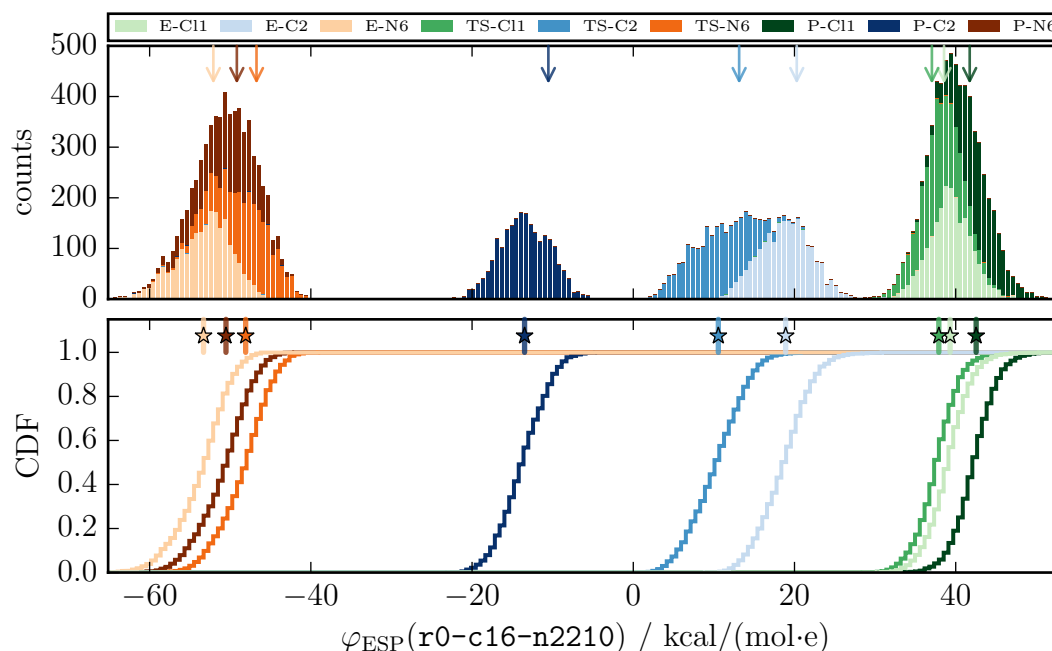


Fig. 10: $N_{\text{Ch}} = 10$ GOCATs (PM7, neutral, COSMO): Stacked histogram and cumulative distribution function of φ_{ESP} for the cluster surrounding the best GOCAT. For illustration details see Fig. 7. The best rank, r_0 , is given in Fig. 11 corresponding to the arrows in the top plot. All average φ_{ESP} values (vertical bars, bottom plot) in $\text{kcal mol}^{-1}\text{e}^{-1}$ for E, TS and P frames (standard deviation in parenthesis): Cl: 39.61(2.69), 37.90(2.65), 42.54(2.67); C: 18.90(3.30), 10.55(3.52), $-13.49(3.06)$; N: $-53.31(3.40)$, $-48.09(3.27)$, $-50.53(3.38)$. These are compiled in Table S4 (ESI).

A histogram of the ESP values of this cluster is given in Fig. 10. Compared to the previous case (Fig. 7 in Section 3.3), the most striking differences and commonalities can be summarized as follows:

1. The overall ESP range across all GOCAT clusters is increased strongly, from $\sim[-24, +24]$ to $\sim[-84, +84] \text{ kcal mol}^{-1}\text{e}^{-1}$.
2. N is now more negatively embedded than C (in the previous Section, C had the most negative embedding).

3. C still shows the biggest ESP shift, but now between the TS and P frames. Before, the ESP shift happened between E and TS.

So still, the Cl atom at all frames (E, TS, P) is very positively embedded with small changes between the frames, but now at about $\varphi_{\text{ESP,Cl}} = 40 \text{ kcal mol}^{-1} \text{e}^{-1}$. The shift of the ESP at the C atom starts now at a positive potential at E and shifts to a negative one at P between $\varphi_{\text{ESP,C}} = 19$ and $-13 \text{ kcal mol}^{-1} \text{e}^{-1}$, while the TS frame is also positively embedded at about $\varphi_{\text{ESP,C}} = 11 \text{ kcal mol}^{-1} \text{e}^{-1}$. The ESP at N still is negative, but this time also amplified to a region at about $\varphi_{\text{ESP,N}} = -50 \text{ kcal mol}^{-1} \text{e}^{-1}$.

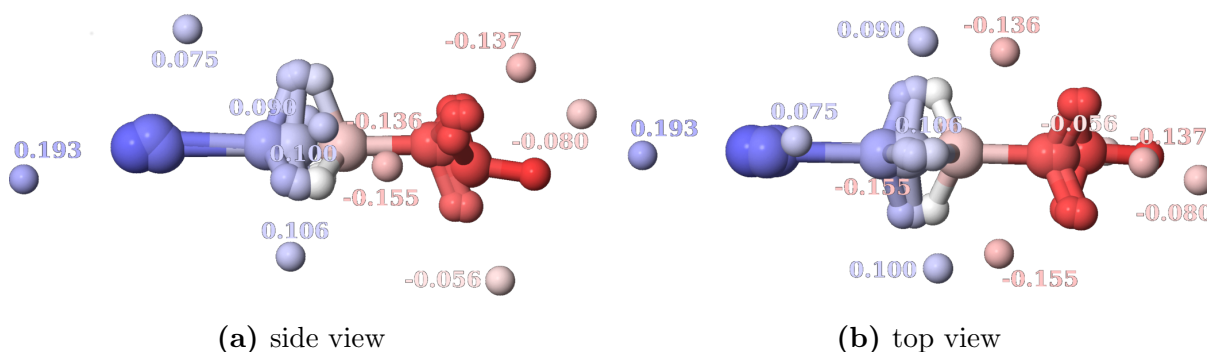


Fig. 11: Single $N_{\text{Ch}} = 10$ GOCAT individual (PM7, neutral, COSMO): r0 (of c16) case with values given for the partial charges. Both partial charges and the atoms of the core frames (E, TS, P shown) are colored red/blue in the ranges $[-0.751, +0.751] \text{e}$ for charges and $[-84.142, +84.142] \text{ kcal mol}^{-1} \text{e}^{-1}$ for ESP values. More views are given in the ESI (Fig. S44).

The amplification part (stronger Coulomb potentials), item 1., can be traced back to the strong stabilizations of the ionic products that are necessary to reach the exothermicity of this reaction path observed in such very polar embeddings. This can also be seen in Fig. 9 (compare the GOCAT profiles to the unoptimized reference energy in that plot): The possible impact of the electrostatic surrounding will always be bigger at the P frame than at the E frame because of the needed charge separation in the contact ion pair. In a crude classical picture: Along with a strong electrostatic stabilization come bigger (additional) gradients that have to be “compensated” by the internal molecular gradients, in order to become the stationary point on the effective PES. In the gas phase, this shift of stationary points had to be suppressed right away (because of the pre-calculated reaction path). Now the shifts

of stationary points of the COSMO path are strictly necessary to be stabilized and the surrounding can or must turn on its influence.

Furthermore, in order to discuss further subtle details, we need to consider the geometric differences between the gas phase and COSMO NEB paths (compare Fig. 2 and Table 1): In the COSMO case, before NH_3 begins its attack onto the middle C atom, it must rotate “inwards”, since within a polar embedding the free electron pair at N is stabilized when pointing outwards. Additionally, and more importantly, the stationary points on the PES are “shifted” with regard to the gas phase: The TS is “earlier” on the reaction coordinate. At the TS, the bond angles are much closer to 90° , i.e., the Walden inversion is not almost finished, as it is in the gas phase TS. Finally, in the COSMO case, the ionic products Cl^- and H_3NCH_3^+ are more widely separated, by about $0.6 \text{ \AA}$ corresponding to a longer summed up reaction coordinate and distance between the TS and P frames. Note that at first sight these structural differences may appear to be small, but they are accompanied by significant gradient changes: By construction, the TS of the COSMO path has a gradient value of zero in the COSMO reference calculation, and it has a small gradient in reasonable GOCATs. However, in the gas phase it has a gradient of $78.5 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$, well beyond the allowed tolerance. Hence, the (necessary) gradient norm penalties prevented the GOCAT optimizations in Section 3.3 to achieve reaction profiles of the COSMO-like kind shown in the present subsection.

In Section 3.3, within the GOCAT case starting with the gas phase NEB path, we saw a boundary of positive and negative charges between Cl and C (see Fig. 8 and text above). The second boundary to the positive partial charges between C and N leads to the result that the biggest (negative) ESP value was found at C (at the TS and P frames). Now within COSMO of this Section, there is just a single boundary of partial charges: Positive charges are sitting either at Cl or besides C (at the E frame and also at the TS frame because of the TS shift), and negative ones at the N side, see Fig. 11, where the very symmetric best rank of the single cluster discussed in this Section is illustrated. Thus, this observation geometrically reflects item 2. and item 3.: The ESP values are positive at the Cl site and monotonically

become negative at the N site. The shift of ESP at the C atom arises between TS and P and not between E and TS, since the TS frame has moved towards the E frame.

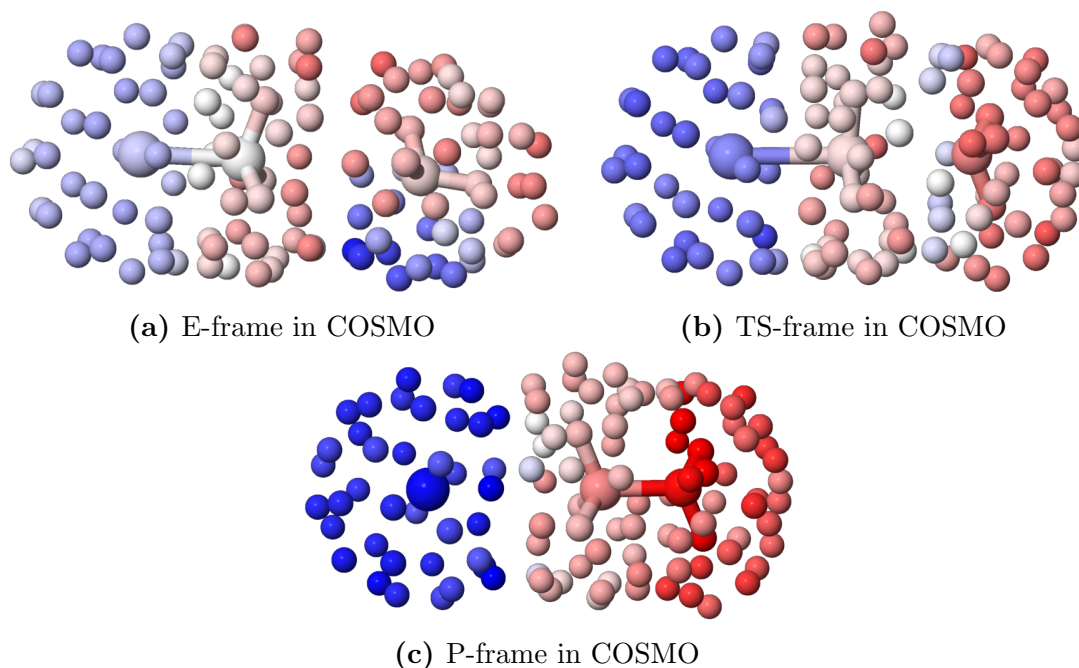


Fig. 12: Side vies of E, TS and P stationary point structures on PM7 with COSMO (H_2O). The quadrature used within COSMO is illustrated by explicit partial charges. The latter are colored red/blue in the range $[-0.03, +0.03]$ e and the φ_{ESP} -values are also calculated and colored in the range $[-92.92, +92.92]$ kcal mol $^{-1}$ e $^{-1}$. Explicit ESP values in kcal mol $^{-1}$ e $^{-1}$ for E, TS and P frames: Cl: 13.95, 38.39, 85.37; C: $-2.83, -4.17, -33.45$; N: $-7.38, -37.21, -82.85$. These are compiled in Table S4 (ESI).

A further illustration of our electrostatic GOCAT embeddings can be obtained by comparison to the COSMO-embedding itself. For this purpose, this COSMO-embedding is translated into a partial charge surrounding, as illustrated in Fig. 12. Per construction in COSMO as apparent surface charge model, partial charges are shown as representatives of the quadrature over the discrete surface tesserae of constant charge density on the cavity's surface.⁵ So, in a perfect conductor the apparent electrostatic potentials at the cavity's surface are exactly compensated by a potential due to the induced (counter-)charges on the cavity's surface that shield (screen) the solute (and afterwards rescaled with a simple screening factor to reach the

⁵ Due to some numerical problems and/or a principally different surface construction mechanism in MOPAC compared to our framework, this surface was constructed on a solvent accessible surface (SAS) with a slightly decreased vdW radius of the probe molecule: This is just for convenience of illustration. The final ESP values as well as COSMO properties were not influenced by this modification.

finite dielectric constant of water).^{111,113}

This pure COSMO-embedding can now serve as reference for the interpretation of our GOCATs. We could think of the cluster of GOCATs treated in this Section and also of the single GOCAT shown in Fig. 11 as globally best and coarsened approximation to COSMO. (Of course, this is for illustrative purposes only, and this comparison ignores several technical details, e.g.: using a vdW surface instead of a solvent accessible surface, or using a minimal distance threshold between the charges instead of near uniformly distributed charges on segments or tesserae.)

Comparing the GOCAT solutions (Fig. 11 and ESP values in Fig. 10) with the pure COSMO case in Fig. 12, we see again ESP values with the trend: $\varphi_{\text{ESP,Cl}} > \varphi_{\text{ESP,C}} > \varphi_{\text{ESP,N}}$ with positive embeddings for Cl, and negative ones for every other atom, including C and N (but also including each separate H atom, compare with Table S4 in the ESI), which was item 2. above. As easily anticipated, during the reaction starting with neutral educts and ending with charged products, the absolute ESP values just increase (the total range of ESP values enlarges), to better screen (and solvate) the products.

The neutral educt frame in COSMO has very small ESP values (and correspondingly small charges), while also some positive (blue) charges can be found close to the free electron pair of NH_3 . Besides the fact of this stabilization (which is also the reason for lifting the C_{3v} symmetry for the educts), this obviously explicitly addresses individual atoms better and is of course due to the SAS construction principle (i.e., almost uniform segment-based embedding which also leads to many, but very small partial charges). Still, the accumulated electrostatic potential at N is negative throughout. On the product side, the charges turn up their impact and their electrostatic potential, to fully stabilize the charged products. In contrast, in the GOCAT model we have just a few charges (with higher absolute values) that try to find a compromise: As the COSMO NEB is fixed and not relaxed for each separate GOCAT in our GA, the common cavity (vdW surface) is the same for each frame and not different for each frame like in the COSMO case. Moreover, a partial charge that tries to stabilize the

product frame a lot (which is the most important handle, of course), will influence the educts, too. In other words, there is just *one* GOCAT surrounding the complete reaction and not a separate one for each NEB frame. So, this does not allow a “dynamic” change or re-action of the surrounding with regard to the changed electrostatic situation of the reaction frames. In contrast, in COSMO it is assumed that the surrounding is instantaneously following the reaction in a frame-wise manner (however small the steps from one frame to the other may be), and that the bidirectional influence between the surroundings and the geometric location of the path has been iterated to self-consistency. The positive charges in the NH_3 educt frame missing in the GOCAT case as well as significantly higher ESP values at the educt is thus one reason for some subtle intermediate minima between E and TS (in some GOCATs, the energy is not strictly monotonically increasing from E to TS, see Fig. 9). Additionally, the overall “compromise” leads to too much stabilization at the TS frame, and to less stabilization at the P frame due to finally smaller absolute φ_{ESP} -values (compare the best rank $\mathbf{r}_0$ with the COSMO reference in Fig. 9): $\Delta E_{\text{E-GOCAT,stab.}} > \Delta E_{\text{E-COSMO,stab.}} \wedge \Delta E_{\text{TS-GOCAT,stab.}} < \Delta E_{\text{TS-COSMO,stab.}} \wedge \Delta E_{\text{P-GOCAT,stab.}} > \Delta E_{\text{P-COSMO,stab.}}$. Finally, the one and only shift of ESP is observed at the C atom, as this exhibits the biggest Cartesian move from E to P (within the maximally compact superposed reaction frames) and can thus be influenced differently, item 3. above.

In summary, keeping the technical differences discussed above in mind, our electrostatic GOCATs can be re-interpreted as coarsened, non-dynamic approximations to COSMO embeddings, for this specific Menshutkin reaction showing this clear charge separation. As shown above, sufficient similarities exist between our best GOCATs and those COSMO embeddings to take the latter as further support that the former work correctly and successfully.

4 Further discussion of the model

The present contribution is a proof-of-concept study, containing several fundamental model approximations, and even after future elimination of these approximations several further steps need to be taken to arrive at a truly comprehensive inverse design of catalysts. To avoid mis-categorizations of our work, these issues are addressed in Section 4.2 below. Before that, in Section 4.1 we briefly digress towards possible connections between different levels of theory.

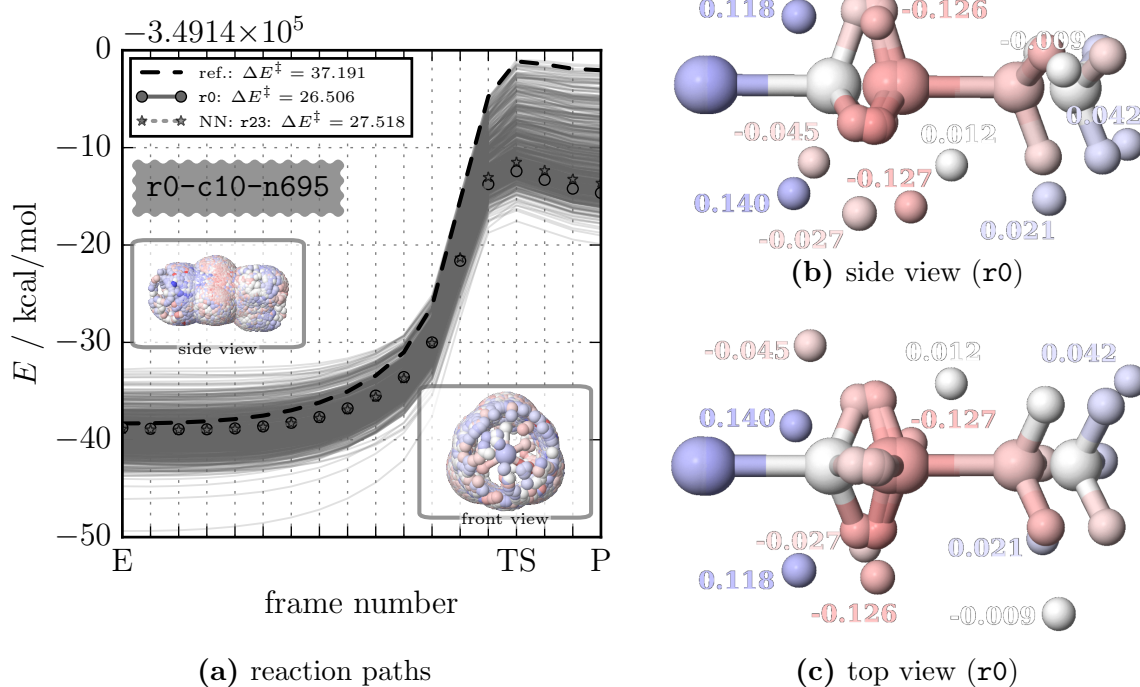


Fig. 13: $N_{\text{Ch}} = 10$ GOCATs (PBE0/def2-TZVP, neutral): In **a**: Reaction paths of the biggest cluster (c10) are shown; other illustration details similar to Fig. 5. In **b**, **c**: Two views on the best GOCAT individual $r0$ in that cluster. Both partial charges and the atoms of the core frames (E, TS, P) are colored red/blue in the ranges $[-0.645, +0.645]$ e for charges and $[-43.675, +43.675]$ kcal mol $^{-1}$ e $^{-1}$ for ESP values, explicitly: Cl: 9.61, 9.94, 9.75; C: 0.97, -10.80, -12.09; N: -0.78, -6.34, -6.47.

4.1 Level of theory

In Section 3, we have deliberately limited ourselves to the PM7 level of theory. This choice was motivated by the possibility to do many exploratory calculations quickly, while still retaining

a quantum-chemical foundation. Of course, however, our GOCAT concept is completely independent of any particular level of theory. For example, we could repeat the present work, using e.g. MP2 or DFT with a suitable functional. This would increase computation times roughly by a factor of 100. However, after having established with the present work how to set up the necessary algorithmic ingredients (e.g., how to choose the fitness function) and by exploiting the built-in trivial parallelism, such a project is now realistic.

To explore possible difficulties and to generate a preliminary glimpse at possible results of such a larger-scale study, we have already done a stripped-down version of it, using PBE0/def2-TZVP in the ORCA program suite.

Note that there are subtle differences in details: For PM7 as implemented in MOPAC, just the φ_{ESP} at the atomic positions are needed;⁸⁵ usually, one would need also a semi-empirical parametrization for the electrostatic potential on the same footing.^{114,115} In contrast, for DFT in ORCA, the full scalar-field Coulomb potential is used during the SCF, as induced by additional “external” point charges (i.e., analogous to new fractional “nuclear core” positions with nuclear attraction/repulsion but no additional bases). Thus, in these DFT calculations, each separate charge *position* is important, not just the final summed Coulomb potential at the core atoms.

Because of the increase of computational time by about two orders of magnitude, we resorted to several time-savers: (1) We used the PM7 GOCATs as initial seeds, instead of random seeds; (2) we only did two separate GA runs, with an order of magnitude less GA iterations; and (3) we locally optimized only part of the GOCATs in the resulting, joined database (not all of them). From these clustered results, we briefly examine only the biggest cluster here.

In Fig. 13 both reaction profiles are shown as well as the best DFT individual optimized so far. Again, also on this higher level of theory, we are able to get significant catalytic effects of almost $\Delta\Delta E^\ddagger = 10 \text{ kcal mol}^{-1}$ with respect to the (DFT) gas-phase profile. Because of the time-savers listed above, some more spread is still visible. But the final best GOCAT found

so far is qualitatively similar to the PM7 result (compare the GOCAT charges and φ_{ESP} values in Fig. 13 with Fig. 8, compiled also in Table S4 (ESI)). Still, the charge values are a bit smaller corresponding also to a smaller ESP at the core atoms.

Besides that, already geometric translations of PM7 GOCATs to PBE0 without any optimization at all lead to a high percentage of GOCATs that also show at least a minor catalytic effect as well as a significantly better starting fitness of the complete database compared to randomly initiated GOCATs.

However, most final separate best GOCATs we found were not as easily interpretable as the ones on PM7, i.e., the strict separation of positive and negative charges in the Cartesian domain and the high symmetry was not (yet) found. We ascribe this to the (prematurely converged) smaller DFT database. Some more impressions on this level of theory are given in the ESI, including also an example of relaxed stationary points within a GOCAT (a posteriori lifting the assumption of a fixed reaction path).

4.2 Model approximations

In Table 2 both some principal model assumptions and important steps are illustrated how the proposed simple electrostatic GOCAT model could be extended to approach a more comprehensive catalysis design process.

Table 2: Overview of important ingredients to reach a better representation of a catalyst (not exhaustive).

Real Catalytic Embedding		Electrostatic GOCAT Model
“relaxed” reaction path	$\Leftrightarrow$	fixed/pre-optimized reaction path
multi-step reaction path	$\Leftrightarrow$	single step (one TS)
catalyst may enable new mechanism	$\Leftrightarrow$	given, fixed mechanism
effects of catalyst-substrate dis-/association	$\Leftrightarrow$	catalyst-substrate complex only
arbitrary substrate-catalyst interactions	$\Leftrightarrow$	QM/MM electrostatic embedding
concrete real catalyst	$\Leftrightarrow$	abstract embedding
ZPE, thermal energy, free enthalpy (including entropy)	$\Leftrightarrow$	potential energy
possible catalyst re-organization	$\Leftrightarrow$	one GOCAT for the whole reaction path

As illustrated in Section 3 by the marked difference between gas-phase-based and COSMO-

based GOCATs, a significant limitation of our GOCAT model in its present, early development stage is keeping the reaction path fixed throughout, both as a path in coordinate space and as reaction profile in energy with (approximately) retained localizations of stationary points. In the real world, it is likely that the catalytic embedding changes the reaction path in both of these respects (i.e., imagine changed mechanisms at an heterogeneous scaffold, within an enzyme or at a transition metal complex; in the context of the present reaction, also an S_N2 to S_N1 shift seems likely). Thus a full reaction-path relaxation within the GOCAT embedding during GOCAT optimization would be advantageous: This would eliminate the need for additional gradient thresholds and would allow for bigger GOCAT impacts. In the examples above, we would have reached COSMO-like GOCATs despite starting out from the gas-phase reaction path. However, a price to pay for this advantage is a substantial increase in computational effort. Work along these lines is in progress.

As already hinted at in the Introduction, many people understand “catalysis” (at transition-metal centers, in enzymes, etc.) as complicated, multi-step reaction sequences, which can be formulated as multi-step reaction cycles in which the catalyst is finally re-generated in its original form. Here we have focused on lowering the activation energy of a single-step reaction. This can either be understood as another (also frequently used) interpretation of “catalysis”, or we can of course argue that concatenation of several (catalyzed) single steps eventually leads to multi-step sequences. Usually, even in longer multi-step sequences a first approach is to narrow down which TS is the actual bottleneck. Then the inverse design problem can be reduced to this step, which again is a single one. For more automation and less user-intervention, frameworks for dynamic reaction networks and automatic optimization of reaction paths exist (for some examples, see Refs. 72–77 and references therein), which could be combined with our GOCAT approach.

However, the influence of a catalyst on a reaction may be even more extreme: It could make new reaction mechanisms possible that are inconceivable in absence of this particular catalyst and hence cannot be part of any pre-conceived single- or multi-step reaction pathway.

Clearly, our present GOCAT setup is very far away from such phenomena. In principle, however, a suitable combination of multistep GOCAT design and automatic reaction network exploration is conceivable for such advanced purposes.

Yet another limitation of our present setup is its exclusive focus on the situation in which catalyst and substrate are in their tightest association. Arguably, this is the most important, central situation of catalysis, but equally clearly it is not the whole story. How the reactants and the catalyst get into this catalytic pose, from an initial infinite separation, and how they dissociate again is also important and has influence on the whole catalytic process, as witnessed e.g. by studies on how proteins “channel” their substrates to, between, and away from their active sites.¹¹⁶

As also hinted at in the Introduction, there is an ongoing controversy on whether catalysis involves various substrate-catalyst interactions (including H-bonds, vdW interactions, etc.) or only electrostatic ones. Our GOCAT model currently represents only the latter but can be extended to also include the former. Indeed, besides partial charges, we aim at introducing other (abstract) interaction groups for also treating vdW interactions, H-bond donors/acceptors, etc.

Of course, a defining feature of our GOCATs is their strong abstraction. As already seen in these very first examples, these abstract catalytic fields can help to analyze and understand known catalytic effects. Similarly, it could help chemists to manually design new, as yet unknown catalysts for a given reaction. However, actually translating a GOCAT into a real catalyst clearly requires substantial further effort. In general, the translation back to a real molecular embedding will likely always introduce some “translation error”, depending on how well chemically realizable the embedding is. An alternative route towards a real chemical catalyst would be a global optimization with real (capped) interaction entities directly from the beginning, i.e., global optimization of a “theozyme” with a more complex objective function than just the TS energy.

Furthermore, real catalysts operate at finite temperature on free enthalpies, and clearly

the association between substrate and catalyst has an effect on entropy. In contrast, our GOCATs currently operate only on bare potential energy values. While adding in some statistical thermodynamics corrections within the usual harmonic approximations is easily possible, taking into account all necessary entropy and anharmonicity effects is challenging.

As noted in Section 3.4, a substantial difference between the COSMO embedding and our GOCAT was that the latter provides one compromise embedding across all points on the reaction path while the former by hypothesis instantly adapts to all changes in the reaction center along the reaction path. It would be possible to extend the GOCAT concept towards multiple GOCATs along the reaction path, with restraints to limit changes between successive ones to what is physically possible. This would then model a GOCAT with conformational freedom and changing polarization states (charges), contributing to the reaction coordinate.

5 Conclusions

Finding an efficient catalyst for a given chemical reaction can be understood as a molecular design task. As a first stage on a longer journey towards this goal, we have presented our “Globally Optimal Catalyst” (GOCAT) framework. It introduces an abstract representation of the catalytic surrounding that is described by a strictly electrostatic model in the present contribution, to reduce and simplify the search space. Within this abstract space, we then search for optimal point charge arrangements, using non-deterministic global search (Genetic Algorithms) and a multicomponent objective function in which “barrier decrease” is only one of several ingredients.

As an illustrative example, we have applied this first-stage framework to the prototype Menshutkin S_N2 reaction, using the semi-empirical PM7 level of theory to enable extensive algorithm testing and varying numbers of point charges (from 1 to 10). Both for the gas-phase and for the solution-phase reaction path, we could find point-charge surroundings with significant catalytic effects. Additionally, our GOCATs for the solution-phase path (but

without a solvent model present) clearly mimic the surrounding that the continuum solvent model provides.

To make the voluminous databases of GOCAT candidates that our GA runs produce accessible to comparative inspection and interpretation, we employed dimension reduction techniques as well as unsupervised Machine Learning (Clustering). For the presented example reaction, our findings indicate that already in its first development stage this GOCAT approach has the potential both to illuminate the decisive interactions behind electrostatic catalytic effects and to aid in finding new catalytic embeddings.

We have concluded this Article with a discussion of further development steps that need to be taken to arrive at the ultimate goal of fully automatic design of optimized real-world catalysts for any given reaction.

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Supporting Information Available

The following supporting information is available: For each case presented in the main article, there are more complementary figures presented and some statistics given in tables. Additionally, a figure set for $N_{\text{Ch}} = 3$ missing in this article is illustrated. Besides that, more information on the used two objective functions is given as well as a GOCAT optimization without a strict TS gradient norm check within the objective function. This material is available free of charge via the Internet at <http://pubs.acs.org/>.

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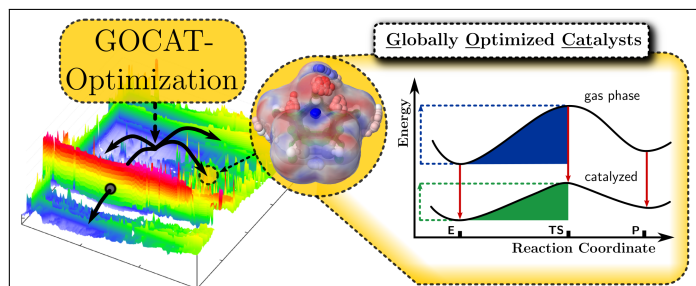
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Graphical TOC Entry



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